

HEATER, VENTILATION & AIR CONDITIONING (HVAC)

07
SECTION

SYMPTOM	CONTROL SYSTEM 07-40
TROUBLESHOOTING 07-03	TECHNICAL DATA 07-50
REFRIGERANT SYSTEM 07-10	SERVICE TOOLS 07-60
BASIC SYSTEM 07-11	

07-03 SYMPTOM TROUBLESHOOTING

FOREWORD 07-03-1	NO.3 NO TEMPERATURE CONTROL
TROUBLESHOOTING INDEX 07-03-1	WITH CLIMATE CONTROL UNIT 07-03-3
NO.1 INSUFFICIENT AIR (OR NO AIR)	NO.4 AIR FROM VENTS COLD
BLOWN FROM VENTS 07-03-1	ENOUGH 07-03-4
NO.2 AMOUNT OF AIR BLOWN FROM	NO.5 NO COOL AIR 07-03-6
VENTS DOES NOT CHANGE 07-03-2	NO.6 NOISE WHILE OPERATING A/C
	SYSTEM 07-03-8

FOREWORD

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- The areas for inspection (steps) are given according to various circuit malfunctions. Use the following chart to verify the symptoms of the trouble in order to diagnose the appropriate area.

TROUBLESHOOTING INDEX

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No.	TROUBLESHOOTING ITEM	DESCRIPTION
1	Insufficient air (or no air) blown from vents.	• Problem with each vent and/or duct.
2	Amount of air blown from vents does not change.	• Malfunction in blower system.
3	No temperature control with climate control unit.	• Malfunction in air outlet duct and/or climate control unit air mix system.
4	Air from vents not cold enough.	• Magnetic clutch operates but A/C system malfunctions.
5	No cool air.	• Magnetic clutch does not operate.
6	Noise while operating A/C system.	• Noise from magnetic clutch, A/C compressor, hose or refrigerant line.

NO.1 INSUFFICIENT AIR (OR NO AIR) BLOWN FROM VENTS

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1	Insufficient air (or no air) blown from vents.
DESCRIPTION	• Problem with each vent and/or duct.
POSSIBLE CAUSE	• Malfunction in VENT mode system (Steps 1—4) • Malfunction in HEAT mode system (Step 5) • Malfunction in DEFROSTER mode system (Steps 6—8)

Diagnostic procedure

Step	Inspection	Inspection item (s)
1	INSPECT TO SEE WHETHER MALFUNCTION IS IN VENT MODE OR OTHER MODES • Does air blow out when in VENT mode?	Yes Go to Step 4. No Go to next step.
2	INSPECT VENT • Is vent clogged?	Yes Remove obstruction, then go to Step 8. No Go to the next step.
3	VERIFY THAT DUCT IN DASHBOARD IS INSTALLED • Is duct in dashboard properly installed?	Yes Inspect duct for clogging, deformity and air leakage, then go to Step 8. No Install duct securely in the proper position, then go to Step 8.

SYMPTOM TROUBLESHOOTING

Step	Inspection	Inspection item (s)
4	INSPECT TO SEE WHETHER MALFUNCTION IS IN HEAT MODE OR DEFROSTER MODE Does air blow out when in HEAT mode?	Yes Go to the next step.
		No Inspect vent for clogging, then go to Step 8.
5	INSPECT DEFROSTER MODE Does air blow out when in DEFROSTER mode?	Yes Operation is normal. Recheck malfunction symptoms.
		No Go to the next step.
6	INSPECT VENT Is vent clogged?	Yes Remove obstruction, then go to Step 8.
		No Go to the next step.
7	VERIFY THAT DEFROSTER DUCT IS INSTALLED Is defroster duct properly installed?	Yes Inspect duct for clogging, deformity, and air leakage, then go to the next step.
		No Install duct securely in proper position, then go to the next step.
8	CONFIRM THAT MALFUNCTION SYMPTOMS DO NOT RECUR AFTER REPAIR Does air blow out?	Yes Troubleshooting completed. Explain repairs to customer.
		No Recheck malfunction symptoms, then repeat from Step 1 if malfunction recurs.

NO.2 AMOUNT OF AIR BLOWN FROM VENTS DOES NOT CHANGE

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2	Amount of air blown from vents does not change.
DESCRIPTION	Malfunction in blower system.
POSSIBLE CAUSE	- Blower relay, blower motor, resistor, fan switch malfunction (Step 1) - Blower unit malfunction (Steps 2—4)

Diagnostic procedure

Step	Inspection	Inspection item (s)
1	INSPECT BLOWER SYSTEM - Inspect the following systems and electrical parts. - Blower relay - Blower motor - Resistor - Fan switch - Related wiring harnesses - Are they normal?	Yes Go to the next step.
		No Repair or replace malfunctioning part, then go to Step 5.
2	CHECK TO SEE WHETHER MALFUNCTION IS IN BLOWER UNIT OR ELSEWHERE - Turn engine switch to ON position. - Turn fan switch on. - Recirculate air inside vehicle. - Does fan in blower unit rotate smoothly?	Yes Go to Step 4.
		No Go to the next step.
3	INSPECT BLOWER UNIT - Inspect fan in blower unit. - Is fan free of interference from blower unit case? - Is fan free of foreign material and obstructions? - Is fan normal?	Yes Go to the next step.
		No Remove obstruction, repair or replace fan and blower unit case, then go to Step 5.
4	INSPECT BLOWER UNIT INTAKE VENT Is blower unit intake vent clogged?	Yes Remove obstruction, then go to the next step.
		No Inspect if there are any obstructions in passage between blower unit and air outlet duct, then go to the next step.
5	VERIFY THAT MALFUNCTION SYMPTOM OCCURS AFTER REPAIR Does air blow out?	Yes Troubleshooting completed. Explain repairs to customer.
		No Recheck malfunction symptoms, then repeat from Step 1 if malfunction recurs.

SYMPTOM TROUBLESHOOTING

NO.3 NO TEMPERATURE CONTROL WITH CLIMATE CONTROL UNIT

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3	No temperature control with climate control unit.
DESCRIPTION	• Malfunction in air outlet duct and/or climate control unit air mix system.
POSSIBLE CAUSE	• Air outlet duct air mix link, air mix crank, air mix rod, air mix wire, wire clamp malfunction (Steps 2, 3) • Climate control unit rack-and-pinion, air mix wire malfunction (Step 4) • Air outlet duct air mix door malfunction (Steps 5, 6) • Heater piping malfunction (Step 7)

Diagnostic procedure

Step	Inspection	Inspection item (s)
1	INSPECT COOLANT TEMPERATURE • Is coolant sufficiently warmed up?	Yes Go to the next step.
		No Warm engine up, then go to Step 8.
2	INSPECT AIR OUTLET DUCT AIR MIX SYSTEM • Inspect air outlet duct air mix links, air mix cranks, air mix rods or wire clamp. — Is there grease on links and cranks? — Are links, cranks, and rods securely installed in their proper positions? — Is wire clamp free of deformation? • Are above items normal?	Yes Go to the next step.
		No Apply grease or install links, cranks, and rods securely in their proper positions, repair or replace wire clamp, then go to Step 8.
3	VERIFY THAT AIR MIX WIRE FROM AIR OUTLET DUCT IS POSITIONED SECURELY AND CORRECTLY (IF AVAILABLE) • Is air mix wire securely installed in the correct position in relation to air outlet duct air mix links?	Yes Go to the next step.
		No Adjust air mix wire or install securely in correct position, then go to Step 8.
4	INSPECT CLIMATE CONTROL UNIT • Inspect climate control unit as follows. — Is bevel gear set properly engaged? — Is air mix wire properly installed in correct position in relation to bevel gear? — Is wire clamp free of deformation? • Are above items okay?	Yes Go to the next step.
		No Properly engage bevel gear or install air mix wire in correct position, repair or replace wire clamp, then go to Step 8.
5	INSPECT AIR OUTLET DUCT • Is there any foreign material or obstruction in air outlet duct air mix doors?	Yes Remove obstruction, then go to Step 8.
		No Go to the next step.
6	INSPECT AIR OUTLET DUCT AIR MIX DOORS • Is air outlet duct air mix door securely and properly installed?	Yes Inspect air mix door for cracks or damage, then go to the next step.
		No Install air mix door securely in proper position, then go to the next step.
7	INSPECT HEATER LINES • Inspect heater lines. — Is heater piping free of damage and cracks? — Are heater piping connections free of engine coolant leakage? — Are heater piping connections securely tightened? — Are heater piping installation points on air outlet duct free of engine coolant leakage? • Are above items normal?	Yes Operation is normal. Recheck malfunction symptoms.
		No If heater piping connections is loosed, tighten connections with specified torque. Repair or replace heater piping, then go to the next step.
8	VERIFY THAT MALFUNCTION SYMPTOM OCCURS AFTER REPAIR • Does unit operate in every temperature setting?	Yes Troubleshooting completed. Explain repairs to customer.
		No Recheck malfunction symptoms, then repeat from Step 1 if malfunction recurs.

SYMPTOM TROUBLESHOOTING

NO.4 AIR FROM VENTS COLD ENOUGH

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4	Air from vents not cold enough.
DESCRIPTION	• Magnetic clutch operates but A/C system malfunctions.
POSSIBLE CAUSE	• Drive belt malfunction (Step 2) • Malfunction in blower unit or condenser (Steps 4, 5) • Malfunction in receiver/drier or expansion valve (valve closes too much) (Steps 8, 9) • Malfunction in refrigerant lines (Steps 10, 11) • A/C compressor system malfunction, insufficient compressor oil (Steps 15, 16) • Over filling of compressor oil, malfunction in expansion valve or air outlet duct air mix link system (Steps 17—19)

Diagnostic procedure

Step	Inspection	Inspection item (s)
1	INSPECT DRIVE BELT • Inspect drive belt. (See 01-10A-3 DRIVE BELT INSPECTION [WL-3].) (See 01-10B-3 DRIVE BELT INSPECTION [WL-C, WE-C].) • Is it normal?	Yes Go to the next step.
		No Adjust or replace drive belt, then go to Step 20.
2	INSPECT REFRIGERANT SYSTEM PERFORMANCE • Perform refrigerant system performance test. (See 07-10-2 REFRIGERANT SYSTEM PERFORMANCE TEST.) • Is operation normal?	Yes Operation is normal. (Recheck malfunction symptoms.)
		No Go to the next step.
3	CHECK TO SEE WHETHER MALFUNCTION IS IN BLOWER UNIT INTAKE AND CONDENSER OR ELSEWHERE • Are refrigerant high-pressure and low-pressure values both high?	Yes Go to the next step.
		No Go to Step 6.
4	INSPECT BLOWER UNIT INTAKE • Is blower unit intake clogged?	Yes Remove obstruction, then go to Step 20. (If air does not reach evaporator within cooling unit, heat exchange does not occur and refrigerant pressure becomes high. Therefore, removal of obstruction is necessary.)
		No Go to the next step.
5	INSPECT CONDENSER • Inspect condenser. (See 07-11-8 CONDENSER INSPECTION.) • Is it normal?	Yes Adjust refrigerant to specified amount, then go to Step 20. (Excessive amount of refrigerant.)
		No Replace condenser, or repair and clean condenser fins, then go to Step 20.
6	CHECK TO SEE WHETHER MALFUNCTION IS IN EXPANSION VALVE, RECEIVER/DRIER AND REFRIGERANT LINES OR ELSEWHERE • Are refrigerant high-pressure and low-pressure values low?	Yes Go to the next step.
		No Go to Step 14.
7	CHECK TO SEE WHETHER MALFUNCTION IS IN EXPANSION VALVE AND RECEIVER/DRIER OR ELSEWHERE • Immediately after A/C compressor operates, does refrigerant high-pressure value momentarily rise to correct value, then fall and stay below it? (Is there negative pressure on low-pressure side?)	Yes Go to the next step.
		No Go to Step 10.
8	CHECK TO SEE WHETHER MALFUNCTION IS IN EXPANSION VALVE OR RECEIVER/DRIER • Turn A/C switch off and let air conditioner stop for 10 min. • Start engine. • Turn both A/C switch and fan switch on. • Does malfunction occur after A/C compressor turns on?	Yes Go to the next step.
		No Replace condenser and vacuum refrigerant line 30 min or more by vacuum pump, add refrigerant to specified level, then go to Step 20. (Since water has intermixed in receiver/drier and it is saturated, replacement is necessary.)

SYMPTOM TROUBLESHOOTING

Step	Inspection	Inspection item (s)
9	VERIFY THAT EXPANSION VALVE HEAT-SENSING TUBE WITHIN COOLING UNIT IS POSITIONED SECURELY AND CORRECTLY Is expansion valve heat-sensing tube within cooling unit securely installed in proper position?	Yes Replace expansion valve, then go to Step 20. (Since valve closes too much, replacement is necessary.)
		No Install heat-sensing tube securely in proper position, then go to Step 20.
10	INSPECT REFRIGERANT LINES - Inspect refrigerant lines. - Is piping free of damage and cracks? - Are piping connections free of oil grime? (Visual inspection) - Are piping connections free of gas leakage? - Are piping installation points on condenser free of gas leakage? - Are piping installation points on receiver/drier free of gas leakage? - Are piping installation points on A/C compressor free of gas leakage? - Are piping installation points on cooling unit free of gas leakage? - Perform gas leak inspection using gas leak tester. - Are above items normal?	Yes Go to the next step.
		No If piping or A/C component (s) are damaged or cracked, replace them. Then go to Step 20. If there is no damage, go to Step 13.
11	INSPECT EVAPORATOR PIPING CONNECTIONS IN COOLING UNIT FOR GAS LEAKAGE Are piping connections for evaporator in cooling unit free of gas leakage?	Yes If the vane makes a noise, add 10 ml {10 cc, 0.34 fl oz} of compressor oil to the A/C compressor. Verify that the noise is no longer heard. Adjust refrigerant to specified amount, then go to Step 20.
		No If piping is damaged or cracked, replace it. Then go to Step 20. If there is no damage, go to the next step.
12	INSPECT EVAPORATOR PIPING CONNECTIONS IN COOLING UNIT FOR LOOSE Are piping connections for evaporator in cooling unit loose?	Yes Tighten connections with specified torque, adjust both compressor oil and refrigerant to specified amount, then go to Step 20.
		No If the vane makes a noise, add 10 ml {10 cc, 0.34 fl oz} of compressor oil to the A/C compressor. Verify that the noise is no longer heard. Replace O-ring on piping, adjust refrigerant to specified amount, then go to Step 20.
13	INSPECT PIPING CONNECTIONS FOR LOOSE Are piping connections loose?	Yes Tighten connections with specified torque, adjust both compressor oil and refrigerant to specified amount, then go to Step 20.
		No If the vane makes a noise, add 10 ml {10 cc, 0.34 fl oz} of compressor oil to the A/C compressor. Verify that the noise is no longer heard. Replace O-ring on piping, adjust refrigerant to specified amount, then go to Step 20.
14	CHECK TO SEE WHETHER MALFUNCTION IS IN EXPANSION VALVE, AIR MIX ACTUATOR AND COMPRESSOR OIL OR ELSEWHERE Does refrigerant high-pressure value hardly increase?	Yes Go to the next step. (Pressure hardly increases.)
		No Go to Step 17.
15	CHECK TO SEE WHETHER MALFUNCTION IS IN COMPRESSOR OIL AMOUNT AND A/C COMPRESSOR OR ELSEWHERE When engine is racing, does high-pressure value increase?	Yes Return to Step 3.
		No Go to the next step.
16	CHECK TO SEE WHETHER MALFUNCTION IS IN COMPRESSOR OIL AMOUNT OR A/C COMPRESSOR After compressor oil is replenished each 10 ml {10 cc, 0.34 fl oz}, does high-pressure value increase?	Yes Troubleshooting completed. (Explain to customer that cause was insufficient compressor oil.)
		No Replace A/C compressor, then go to Step 20. (Cause is defective A/C compressor.)
17	CHECK TO SEE WHETHER MALFUNCTION IS IN EXPANSION VALVE OR ELSEWHERE Is only refrigerant low-pressure value high?	Yes Go to Step 19.
		No Go to the next step.

SYMPTOM TROUBLESHOOTING

Step	Inspection	Inspection item (s)
18	VERIFY THAT AIR MIX IS INSTALLED SECURELY AND PROPERLY Are air outlet duct air mix links, air mix cranks, and air mix rods securely and properly installed?	Yes Set fan switch to 4th position. Turn A/C switch on. Set FRESH mode. Set temperature control to MAX COLD. Set VENT mode. (1) Start and run the engine at 1,500 rpm for 10 min. (2) Run the engine at idle for 1 min. (3) Within 12 s , idle → 4,000 rpm → idle. Perform cycle 5 times. (4) Run the engine at idle for 30 s. (5) Drain the compressor oil completely from the A/C compressor and verify the amount. - If there is approx. 90 ml {90 cc, 3.0 fl oz} of compressor oil, go to Step 20. - If there is more than 90 ml {90 cc, 3.0 fl oz} of compressor oil, remove surplus oil and fill A/C compressor with 90 ml {90 cc, 3.0 fl oz} of compressor oil. Repeat Steps (1) to (5). (Cause is excessive amount of compressor oil.)
		No Repair or install links, cranks and rods securely in proper position, then go to Step 20.
19	VERIFY THAT EXPANSION VALVE HEAT-SENSING TUBE WITHIN COOLING UNIT IS POSITIONED SECURELY AND CORRECTLY Is expansion valve heat-sensing tube within cooling unit securely installed in proper position?	Yes Replace expansion valve, then go to the next step. (Since valve opens too much, replacement is necessary.)
		No Install heat-sensing tube securely in proper position, then go to the next step.
20	VERIFY THAT MALFUNCTION SYMPTOM OCCURS AFTER REPAIR Does cool air blow out? (Are results of refrigerant system performance test normal?)	Yes Troubleshooting completed. Explain repairs to customer.
		No Recheck malfunction symptoms, then repeat from Step 1 if malfunction recurs.

NO.5 NO COOL AIR

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5	No cool air
DESCRIPTION	Magnetic clutch does not operate.
POSSIBLE CAUSE	- Malfunction in PCM A/C cut control system (Step 3) - Malfunction in climate control unit (Step 4) - Malfunction in refrigerant pressure switch (Steps 5, 7—9) - Malfunction in PCM (A/C signal) (Step 6) - Malfunction in PCM (IG1 signal) (Steps 10, 11) - Malfunction in A/C compressor (Step 12) - Malfunction in A/C relay (Steps 13—15) - Malfunction in evaporator temperature sensor (Step 16)

- When performing an asterisked (*) troubleshooting inspection, shake the wiring harness and connectors while performing the inspection to discover whether poor contact points are the cause of any intermittent malfunctions. If there is a problem, check to make sure connectors, terminals and wiring harnesses are connected correctly and undamaged.

Diagnostic procedure

Step	Inspection	Inspection item (s)
1	INSPECT AIR BLOW OUT Does air blow out?	Yes Go to the next step.
		No Go to Step 1 of troubleshooting index No.1.
2	INSPECT A/C COMPRESSOR OPERATION - Start engine. - Turn A/C switch and fan switch on. - Does A/C compressor operate?	Yes Go to Step 1 of troubleshooting index No.4.
		No Go to the next step.
3	INSPECT FOR DTC IN PCM - Inspect for DTCs related to the PCM on-board diagnostic system. - Are any DTCs displayed?	Yes Go to appropriate inspection procedure.
		No Go to the next step.

SYMPTOM TROUBLESHOOTING

Step	Inspection	Inspection item (s)	
4	INSPECT TO SEE WHETHER MALFUNCTION IS IN CLIMATE CONTROL UNIT OR ELSEWHERE Does cool air blow out when terminal 3B of climate control unit connector (3-pin, A/C signal) is grounded?	Yes	Replace climate control unit, then go to Step 17.
		No	Release short, then go to the next step.
5*	INSPECT TO SEE WHETHER MALFUNCTION IS IN A/C SIGNAL CIRCUIT (BETWEEN REFRIGERANT PRESSURE SWITCH AND PCM) OR ELSEWHERE - Test voltage at following terminal of refrigerant pressure switch. — Terminal B (A/C signal) - Is voltage approx. 12 V?	Yes	Go to Step 7.
		No	Go to the next step.
6*	INSPECT TO SEE WHETHER MALFUNCTION (LACK OF CONTINUITY) IS IN WIRING HARNESS (BETWEEN REFRIGERANT PRESSURE SWITCH AND PCM) OR PCM - Test voltage at A/C signal terminal of PCM. - Is voltage approx. 12 V?	Yes	Repair wiring harness between PCM and refrigerant pressure switch, then go to Step 17.
		No	Inspect PCM, then go to Step 17.
7	INSPECT TO SEE WHETHER MALFUNCTION IS IN REFRIGERANT PRESSURE SWITCH, REFRIGERANT AMOUNT, OR ELSEWHERE Does cool air blow out when terminals A and B of refrigerant pressure switch connector are shorted?	Yes	Go to Step 9.
		No	Go to the next step.
8*	INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (BETWEEN REFRIGERANT PRESSURE SWITCH AND CLIMATE CONTROL UNIT) OR ELSEWHERE - Test voltage at following terminal of climate control unit. (3-pin) — Terminal 3B (3-pin, A/C signal) - Is voltage approx. 12 V?	Yes	Go to Step 10.
		No	Repair wiring harness between refrigerant pressure switch and climate control unit, then go to Step 17.
9	INSPECT TO SEE WHETHER MALFUNCTION IS IN REFRIGERANT PRESSURE SWITCH OR REFRIGERANT AMOUNT - Inspect refrigerant pressure switch. - Is it normal?	Yes	If refrigerant amount is empty, replace condenser, vacuum refrigerant line 30 min or more by vacuum pump, and add refrigerant to specified level, then go to Step 17.
		No	Replace refrigerant pressure switch, then go to Step 17.
10	INSPECT TO SEE WHETHER MALFUNCTION (LACK OF CONTINUITY) IS IN A/C CONTROL SIGNAL CIRCUIT (BETWEEN A/C RELAY AND PCM) OR ELSEWHERE Does cool air blow out when terminal E of A/C relay connector (A/C control signal) is grounded?	Yes	Release short, then go to the next step.
		No	Go to Step 12.
11*	INSPECT TO SEE WHETHER MALFUNCTION (LACK OF CONTINUITY) IS IN PCM OR WIRING HARNESS (BETWEEN A/C RELAY AND PCM) - Test voltage at the A/C relay control signal terminal of PCM. - Is voltage approx. 12 V?	Yes	Inspect PCM, then go to Step 17.
		No	Repair wiring harness between A/C relay and PCM, then go to Step 17.
12*	INSPECT TO SEE WHETHER MALFUNCTION IS IN MAGNETIC CLUTCH OR ELSEWHERE - Test voltage at the following terminal of magnetic clutch. — Terminal A (magnetic clutch operation signal) - Is voltage approx. 12 V?	Yes	Inspect magnetic clutch, then go to Step 17.
		No	Go to the next step.
13	INSPECT FUSE Are A/C relay power supply fuses normal?	Yes	Go to the next step.
		No	Replace fuse, then go to Step 17. If fuse burns out immediately, go to the next step.

SYMPTOM TROUBLESHOOTING

Step	Inspection	Inspection item (s)
14	INSPECT WIRING HARNESS BETWEEN FUSE BLOCK AND A/C RELAY FOR LACK OF CONTINUITY - Test voltages at following terminals of A/C relay. - Terminal A (A/C relay control signal) - Terminal C (A/C control signal) - Are voltages approx. 12 V?	Yes Go to the next step.
		No Repair wiring harness between fuse block and A/C relay, then go to Step 17.
15	INSPECT TO SEE WHETHER MALFUNCTION IS IN A/C RELAY OR WIRING HARNESS (BETWEEN A/C RELAY AND MAGNETIC CLUTCH) AND EVAPORATOR TEMPERATURE SENSOR - Test voltage at the following terminal of A/C relay. - Terminal D (magnetic clutch operation signal) - Is voltage approx. 12 V?	Yes Inspect wiring harness between A/C relay and magnetic clutch. - If above wiring harness is normal, go to the next step. - If above wiring harness malfunctions, repair wiring harness, then go to Step 17.
		No Replace A/C relay, then go to Step 17.
16	INSPECT EVAPORATOR TEMPERATURE SENSOR - Inspect evaporator temperature sensor. - Is it normal?	Yes Go to the next step.
		No Replace evaporator temperature sensor, then go to the next step.
17	CONFIRM THAT MALFUNCTION SYMPTOMS DO NOT RECUR AFTER REPAIR Does cool air blow out? (Are the results of refrigerant system performance test normal?)	Yes Troubleshooting completed. Explain repairs to customer.
		No Recheck malfunction symptoms, then repeat from Step 1 if malfunction recurs.

SYMPTOM TROUBLESHOOTING

NO.7 NO COOL AIR.[5R55S]

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7	No cool air
DESCRIPTION	? Magnetic clutch does not operate.
POSSIBLE CAUSE	? A/C compressor system malfunction (Step 2) ? Incorrect amount of refrigerant (Step 3) ? A/C switch indicator light malfunction (Steps 4—6) ? PCM A/C cut-off control system (WL, WLT-1, WLT-2: Step 18) (G6: Step 7) ? PCM (IG1 signal) system malfunction (WL, WLT-1, WLT-2: Steps 8, 9) (G6: Step 20) ? Thermoswitch, A/C switch malfunction (WL, WLT-1, WLT-2: Steps 11, 12, 14, 15) (G6: Step 10—14) ? PCM (A/C signal) system malfunction (WL, WLT-1, WLT-2: Steps 17) (G6: Step 16, 17) ? Refrigerant pressure switch malfunction (WL, WLT-1, WLT-2: Step 26) (G6: Step 19) ? A/C relay malfunction (WL, WLT-1, WLT-2: Steps 22—25) (G6: Step 22—24)

? When performing an asterisked (*) troubleshooting inspection, shake the wiring harness and connectors while doing the inspection to discover whether poor contact points are the cause of any intermittent malfunctions. If there is a problem, check to make sure connectors, terminals and wiring harnesses are connected correctly and undamaged.

Diagnostic procedure

STEP	INSPECTION	ACTION
1	INSPECT AIR BLOW OUT ? Does air blow out?	Yes Go to next step.
		No Go to Step 1 of troubleshooting indexes No.1, 2.
2	INSPECT A/C COMPRESOR OPERATION ? Start engine. ? Turn both A/C switch and front fan switch on. ? Does A/C compressor operate?	Yes Go to Step 1 of troubleshooting index No. 6.
		No Go to next step.
3	INSPECT REFRIGERANT AMOUNT ? Inspect refrigerant amount. ? Is it okay?	Yes Go to next step.
		No Add or subtract refrigerant to specified level, then go to Step 27.
4	INSPECT A/C SWITCH INDICATOR LIGHT ? Does A/C switch indicator light illuminate?	Yes WL or WL-T1, WLT-2: Go to Step 8. G6: Go to Step 7.
		No Go to next step.
*5	INSPECT WIRING HARNESS BETWEEN FUSE BLOCK AND FRONT CLIMATE CONTROL UNIT FOR CONTINUITY ? Turn ignition switch to ON position. ? Turn both A/C switch and front fan switch on. ? Remove airflow mode wire and air mix wire and pull out front climate control unit. ? Test voltage at front climate control unit terminal C (IG2 signal). ? Is voltage approximately 12 V ?	Yes Go to next step.
		No Repair wiring harness between fuse block and A/C switch, then go to Step 27.
*6	INSPECT A/C SWITCH ? Inspect A/C switch. ? Is it okay?	Yes Inspect wiring harness between A/C switch and resistor, then go to Step 27.
		No Replace A/C switch, then go to Step 27.
7	CHECK FOR DTCS IN PCM ? Check the DTC for the ON-BOARD DIAGNOSTIC (ENGINE CONTROL). ? Are the following DTCs displayed? — DTC 0120	Yes Go to appropriate inspection procedure. (The throttle position sensor (DTC 0120), which sends the PCM's A/C cut off control input signal, may be the cause of the trouble.)
		No Go to Step 10.
*8	VERIFY WHETHER MALFUNCTION IS IN MAGNETIC CLUTCH SYSTEM OR ELSEWHERE ? Turn ignition switch to ON position. ? Does magnetic clutch operate when A/C relay connector terminal B (IG2 signal) is grounded?	Yes Go to next step.
		No Go to Step 21.
*9	VERIFY WHETHER MALFUNCTION IS IN A/C RELAY SYSTEM (COIL-SIDE) OR ELSEWHERE ? Turn A/C switch off. ? Test voltage at PCM connector terminal L(IG1 signal). ? Is voltage approximately 12 V ?	Yes Go to Step 11.
		No Repair wiring harness between A/C relay and PCM, then go to Step 27.

07

SYMPTOM TROUBLESHOOTING

STEP	INSPECTION	ACTION	
*10	VERIFY WHETHER MALFUNCTION IS IN THE MOSWITCH SYSTEM OR ELSEWHERE ? Turn ignition switch to LOCK position. ? Remove radiator grille. ? Disconnect refrigerant pressure switch connector. ? Turn ignition switch to ON position. ? Set front fan switch at first speed. ? Test voltage at the following refrigerant pressure switch connector terminal B (A/C signal) on wiring harness side. ? Is voltage approximately 12 V when A/C switch is off and 0 V when it is on?	Yes	Go to Step 16.
		No	Reconnect refrigerant pressure switch connector, then go to next step.
*11	VERIFY WHETHER MALFUNCTION IS IN EVAPORATOR TEMPERATURE SENSOR AND WIRING HARNESS (BETWEEN FUSE BLOCK AND EVAPORATOR TEMPERATURE SENSOR FOR CONTINUITY) OR ELSEWHERE ? Turn ignition switch to LOCK position. ? Remove glove compartment. ? Disconnect evaporator temperature sensor connector. ? Start engine. ? Turn both A/C switch and front fan switch on. ? When evaporator temperature sensor connector terminals B and C (on wiring harness side) are shorted, does cool air blow out?	Yes	Undo short, then go to next step.
		No	WL or WL-T1, WLT-2: Undo short, then go to Step 14. G6: Undo short, then go to Step 13.
*12	VERIFY WHETHER MALFUNCTION IS IN EVAPORATOR TEMPERATURE SENSOR OR WIRING HARNESS (BETWEEN FUSE BLOCK AND THE MOSWITCH FOR CONTINUITY) ? Turn ignition switch to ON position. ? Test voltage at evaporator temperature sensor connector terminal A (IG2 signal). ? Is voltage approximately 12 V ?	Yes	Inspect evaporator temperature sensor, then go to Step 27.
		No	Repair wiring harness between fuse block and evaporator temperature sensor, then go to Step 27.
*13	INSPECT WIRING HARNESS BETWEEN EVAPORATOR TEMPERATURE SENSOR AND REFRIGERANT PRESSURE SWITCH FOR CONTINUITY ? Turn ignition switch to LOCK position. ? Inspect for continuity between evaporator temperature sensor connector terminal B (A/C signal) and refrigerant pressure switch connector terminal B. ? Is there continuity?	Yes	Go to next step.
		No	Repair wiring harness between refrigerant pressure switch and evaporator temperature sensor, then go to Step 27.
*14	VERIFY WHETHER MALFUNCTION IS IN EVAPORATOR TEMPERATURE SENSOR AND WIRING HARNESS (BETWEEN EVAPORATOR TEMPERATURE SENSOR AND A/C SWITCH FOR CONTINUITY) OR ELSEWHERE ? Turn ignition switch to ON position. ? Turn A/C switch ON. ? Turn fan switch ON. ? Test voltage at A/C switch connector terminal A (A/C signal). ? Is voltage approximately 12 V ?	Yes	WL or WL-T1, WLT-2: Go to next step. G6: Inspect wiring harness between evaporator temperature sensor and A/C switch, then go to Step 27.
		No	Inspect A/C switch, then go to Step 27.

SYMPTOM TROUBLESHOOTING

STEP	INSPECTION	ACTION	
*15	VERIFY WHETHER MALFUNCTION IS IN EVAPORATOR TEMPERATURE SENSOR AND WIRING HARNESS (BETWEEN EVAPORATOR TEMPERATURE SENSOR AND A/C SWITCH FOR CONTINUITY) OR ELSEWHERE ? Test voltage at evaporator temperature sensor connector terminal C (A/C signal). ? Is voltage approximately 12 V ?	Yes	Go to Step 17.
		No	Repair wiring harness between evaporator temperature sensor and A/C switch, then go to Step 27.
*16	VERIFY WHETHER MALFUNCTION IS IN PCM AND WIRING HARNESS (BETWEEN PCM AND REFRIGERANT PRESSURE SWITCH FOR CONTINUITY) OR ELSEWHERE ? Test voltage at refrigerant pressure switch connector (on wiring harness side) terminal A (A/C signal). ? Is voltage approximately 12 V ?	Yes	Go to Step 19.
		No	Go to next step.
*17	VERIFY WHETHER MALFUNCTION IS IN PCM OR WIRING HARNESS (BETWEEN PCM AND REFRIGERANT PRESSURE SWITCH FOR CONTINUITY) ? Test voltage at the following PCM connector terminals (A/C signal). — Terminal G (WL or WL-T1, WLT-2) — Terminal Q (G6) ? Is voltage approximately 12 V ?	Yes	WL or WL-T1, WLT-2: Go to next step. G6: Inspect wiring harness between PCM and refrigerant pressure switch, then go to Step 27.
		No	Inspect PCM, then go to Step 27.
*18	VERIFY WHETHER MALFUNCTION IS IN PCM AND WIRING HARNESS (BETWEEN PCM AND EVAPORATOR TEMPERATURE SENSOR FOR CONTINUITY) OR ELSEWHERE ? Inspect the following input signal components: — Engine coolant temperature sensor, including PCM wiring harness (A/C cut-off control) ? Are they okay?	Yes	Repair wiring harness between PCM and evaporator temperature sensor, then go to Step 27.
		No	Replace input signal components, then go to Step 27.
*19	VERIFY WHETHER MALFUNCTION IS IN REFRIGERANT PRESSURE SWITCH OR ELSEWHERE ? When the following refrigerant pressure switch connector terminal A and B (on wiring harness side) are shorted, does cool air blow out?	Yes	Inspect refrigerant pressure switch, then go to Step 27.
		No	Undo short, reconnect refrigerant pressure switch, then go to next step.
*20	INSPECT TO SEE WHETHER MALFUNCTION IS IN A/C RELAY OR ELSEWHERE ? Does magnetic clutch operate when terminal B (IG2 signal) of A/C relay connector is grounded?	Yes	Inspect wiring harness between A/C relay and PCM, then go to Step 27.
		No	Go to next step.
*21	VERIFY WHETHER MALFUNCTION IS IN MAGNETIC CLUTCH OR WIRING HARNESS (BETWEEN A/C RELAY AND MAGNETIC CLUTCH FOR CONTINUITY) ? Test voltage at magnetic clutch connector terminal A (A/C control signal). ? Is voltage approximately 12 V ?	Yes	Inspect magnetic clutch, then go to Step 27.
		No	Go to next step.
22	INSPECT A/C RELAY (SWITCH-SIDE) POWER SUPPLY FUSE ? Are A/C relay power supply fuses okay?	Yes	Go to next step.
		No	Inspect for a short to ground on blown fuses's circuit. Repair or replace as necessary. Install appropriate amperage fuse.

SYMPTOM TROUBLESHOOTING

STEP	INSPECTION	ACTION	
*23	INSPECT WIRING HARNESS BETWEEN FUSE BLOCK AND A/C RELAY (SWITCH-SIDE) FOR CONTINUITY ? Turn ignition switch to ON position. ? Test voltage at the following A/C relay connector terminals: — Terminal A (IG2 signal) — Terminal C (A/C control signal) ? Is voltage approximately 12 V ?	Yes	Go to next step.
		No	Repair wiring harness between fuse block and A/C relay, then go to Step 27.
*24	VERIFY WHETHER MALFUNCTION IS IN MAGNETIC CLUTCH OR WIRING HARNESS (BETWEEN A/C RELAY AND MAGNETIC CLUTCH FOR CONTINUITY) ? Test voltage at A/C relay connector terminal D (A/C control signal). ? Is voltage approximately 12 V ?	Yes	WL or WL-T1, WLT-2: Go to next step. G6: Repair wiring harness between A/C relay and magnetic clutch, then go to Step 27.
		No	Inspect A/C relay, then go to Step 27.
*25	INSPECT WIRING HARNESS BETWEEN FUSE BLOCK AND A/C RELAY (SWITCH-SIDE) FOR CONTINUITY ? Turn ignition switch to ON position. ? Test voltage at refrigerant pressure switch connector (on wiring harness side) terminal A (A/C control signal) ? Is voltage approximately 12 V ?	Yes	Go to next step.
		No	Repair wiring harness between A/C relay and refrigerant pressure switch, then go to Step 27.
*26	VERIFY WHETHER MALFUNCTION IS IN REFRIGERANT PRESSURE SWITCH OR ELSEWHERE ? When refrigerant pressure switch connector terminal A and B (on wiring harness side) are shorted, does cool air blow out?	Yes	Inspect refrigerant pressure switch, then go to next step.
		No	Repair wiring harness between refrigerant pressure switch and magnetic clutch, then go to next step.
27	VERIFY THAT MALFUNCTION SYMPTOM OCCURS AFTER REPAIR ? Does cool air blow out? (Is refrigerant system performance test result correct?)	Yes	Troubleshooting completed. Explain repairs to customer.
		No	Recheck malfunction symptoms, then repeat from Step 1 if malfunction recurs.

SYMPTOM TROUBLESHOOTING

NO.6 NOISE WHILE OPERATING A/C SYSTEM

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6	Noise while operating A/C system.
DESCRIPTION	Noise from magnetic clutch, A/C compressor, hose or refrigerant line.
POSSIBLE CAUSE	- Magnetic clutch operation noise (Step 4) - A/C compressor vane noise (Steps 5—13) - A/C compressor slippage noise (Steps 14—17) - Hose or refrigerant line interference noise (Step 18)

Diagnostic procedure

Step	Inspection	Inspection item (s)
1	CHECK A/C COMPRESSOR VANE NOISE Is there a jingling, popping, beeping, or buzzing sound (A/C compressor vane noise)?	Yes Go to Step 5.
		No Go to the next step.
2	INSPECT A/C COMPRESSOR SLIPPAGE NOISE Is there a squeaking or whirling sound (A/C compressor slippage noise)?	Yes Go to Step 14.
		No Go to the next step.
3	INSPECT A/C COMPRESSOR INTERFERENCE NOISE Is there a rattling or vibrating sound (interference noise)?	Yes Go to Step 18.
		No Go to the next step.
4	INSPECT MAGNETIC CLUTCH OPERATION NOISE Is there a clicking sound (magnetic clutch operation noise)?	Yes Adjust clearance between pressure plate of magnetic clutch and A/C compressor pulley, then go to Step 19. (See 07-40-4 MAGNETIC CLUTCH ADJUSTMENT.)
		No Condition is normal. (Recheck malfunction symptoms.)
5	INSPECT A/C COMPRESSOR NOISE TIME Is noise heard continuously for 3 s or more after A/C compressor comes on?	Yes Go to the next step.
		No Condition is normal. (Noise occurs for 2—3 s immediately after A/C compressor turns on.)
6	INSPECT IDLE SPEED - Inspect idle speed. (See 01-10A-20 ENGINE TUNE-UP [WL-3].) (See 01-10B-21 ENGINE TUNE-UP [WL-C, WE-C].) - Is it normal?	Yes Go to the next step.
		No Follow the repair instruction described in section F, then go to Step 19.

SYMPTOM TROUBLESHOOTING

Step	Inspection	Inspection item (s)	
7	INSPECT REFRIGERANT AMOUNT - Inspect refrigerant amount. - Is it normal?	Yes	Go to Step 10.
		No	Go to the next step.
8	INSPECT REFRIGERANT LINES - Inspect refrigerant lines. - Is piping free of damage and cracks? - Are piping connections free of oil grime? (Visual inspection) - Are piping connections free of gas leakage? - Are piping installation points on condenser free of gas leakage? - Are piping installation points on receiver/drier free of gas leakage? - Are piping installation points on A/C compressor free of gas leakage? - Are piping installation points on cooling unit free of gas leakage? - Perform gas leak inspection using gas leak tester. - Are above items normal?	Yes	Go to the next step.
		No	If piping or A/C component (s) is damaged or cracked, replace then go to Step 19. If there is gas leakage, repair or replace connection and replace condenser*, then go to Step 19.
9	INSPECT EVAPORATOR PIPING CONNECTIONS IN COOLING UNIT FOR GAS LEAKAGE Are piping connections for evaporator in cooling unit free of gas leakage?	Yes	Adjust refrigerant amount to specified level, then go to Step 19.
		No	If piping is damaged or cracked, replace then go to Step 19. If there is gas leakage, repair or replace connection and replace condenser*, then go to Step 19.
10	CHECK TO SEE WHETHER MALFUNCTION IS IN COMPRESSOR OIL OR ELSEWHERE - Add 20 ml {20 cc, 0.8 fl oz} of compressor oil. - Is noise heard when racing engine?	Yes	Go to the next step.
		No	Troubleshooting completed. Explain repair to customer.
11	CHECK TO SEE WHETHER MALFUNCTION IS IN A/C COMPRESSOR OR ELSEWHERE - Drain compressor oil. - Is it contaminated with metal particles?	Yes	Go to the next step.
		No	Replace A/C compressor, then go to Step 19.
12	CHECK TO SEE WHETHER MALFUNCTION IS SOMEWHERE IN A/C SYSTEM OR ELSEWHERE Is compressor oil whitish and mixed with water?	Yes	Replace entire A/C system (excluding heater), then go to Step 19.
		No	Go to the next step.
13	INSPECT A/C COMPRESSOR OIL Is compressor oil darker than normal and contaminated with aluminum chips?	Yes	Replace A/C compressor and condenser, then go to Step 19. (Since A/C compressor may be worn and receiver/drier may be clogged, replacement of receiver/drier is necessary.)
		No	Condition is normal. Recheck malfunction symptoms.
14	CHECK TO SEE WHETHER MALFUNCTION IS IN A/C COMPRESSOR OR ELSEWHERE Is noise heard immediately after A/C compressor is stopped?	Yes	Replace A/C compressor, then go to Step 19. (A/C compressor discharge valve left open)
		No	Go to the next step.
15	INSPECT DRIVE BELT - Inspect drive belt. (See 01-10A-3 DRIVE BELT INSPECTION [WL-3].) (See 01-10B-3 DRIVE BELT INSPECTION [WL-C, WE-C].) - Is it normal?	Yes	Go to the next step.
		No	Adjust or replace drive belt, then go to Step 19.
16	INSPECT DRIVE BELT CONDITION - Is drive belt worn? - Does it have foreign material imbedded in it, or have oil on it?	Yes	Remove obstruction, remove oil, or replace drive belt, then go to Step 19.
		No	Go to the next step.
17	INSPECT MAGNETIC CLUTCH - Inspect magnetic clutch. (See 07-40-5 MAGNETIC CLUTCH INSPECTION.) - Is it normal?	Yes	Replace A/C compressor (excluding pressure plate, A/C compressor pulley, and stator), then go to Step 19.
		No	Replace magnetic clutch, then go to Step 19.

SYMPTOM TROUBLESHOOTING

Step	Inspection	Inspection item (s)
18	CHECK TO SEE WHETHER MALFUNCTION IS IN A/C COMPRESSOR OR REFRIGERANT LINES Is noise emitted from A/C compressor?	Yes Visually inspect A/C compressor, replace appropriate parts if necessary, then go to the next step.
		No If noise is due to refrigerant lines, repair detached or missing clips, tighten loose bolts, then go to the next step.
19	VERIFY THAT MALFUNCTION SYMPTOM OCCURS AFTER REPAIR Has A/C compressor noise stopped?	Yes Troubleshooting completed. Explain repairs to customer.
		No Recheck malfunction symptoms, then repeat from Step 1 if malfunction recurs.

* : If there is gas leakage, air enters into the A/C system. The desiccant within the receiver/drier absorbs the moisture from the air and becomes saturated. If the A/C system is used in this condition, the inside of the A/C compressor will begin to rust due to this moisture, which may cause lock up or noise to occur. Therefore, replacement of the receiver/drier is necessary.

07-10 REFRIGERANT SYSTEM

REFRIGERANT SYSTEM SERVICE	
WARNINGS	07-10-1
REFRIGERANT SYSTEM SERVICE	
CAUTIONS	07-10-1
REFRIGERANT SYSTEM GENERAL	
PROCEDURES	07-10-2

REFRIGERANT SYSTEM	
PERFORMANCE TEST	07-10-2
REFRIGERANT PRESSURE	
CHECK	07-10-3
RECOVERY	07-10-3
REFRIGERANT CHARGING	07-10-3

REFRIGERANT SYSTEM SERVICE WARNINGS

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Handling Refrigerant

- Avoid breathing air conditioning refrigerant or lubricant vapor. Exposure may irritate eyes, nose and throat. Also, due to environmental concerns, the use of recovery/recycling/recharging equipment is mandatory when draining R-134a from the air conditioning system. If accidental system discharge occurs, ventilate work area before resuming service.
- Do not perform pressure test or leak test for R-134a service equipment and/or vehicle air conditioning system using compressed air. Some mixtures of air and R-134a have been shown to be combustible at elevated pressures. These mixtures, if ignited, may cause injury or property damage. Additional health and safety information may be obtained from refrigerant manufacturers.
- Do not allow the refrigerant to leak near fire or any kind of heat. A poisonous gas may be generated if the refrigerant gas contacts fire or heat such as from cigarettes and heaters. When carrying out any operation that can cause refrigerant leakage, extinguish or remove the above-mentioned heat sources and maintain adequate ventilation.
- Handling liquid refrigerant is dangerous. A drop of it on the skin can result in localized frostbite. When handling the refrigerant, wear gloves and safety goggles. If refrigerant splashes into the eyes, immediately wash them with clean water and consult a doctor.

Storing Refrigerant

- The refrigerant container is highly pressurized. If it is subjected to high heat, it could explode, scattering metal fragments and liquid refrigerant that can seriously injure you. Store the refrigerant at temperatures below 40 °C {104 °F}.

REFRIGERANT SYSTEM SERVICE CAUTIONS

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Handling Insufficient Refrigerant Level

- If an insufficient refrigerant level is detected at troubleshooting, do not charge (add) the refrigerant. Because an accurate amount of refrigerant cannot be determined from the pressure indicated on the manifold gauge, never charge the refrigerant. If there is too much or too little refrigerant from the refilling, there may be secondary problems such as damage to the refrigerant cycle parts, or a decrease of cooling performance. Therefore, if it is determined that the refrigerant level is insufficient, completely remove refrigerant from the refrigerant cycle and refill with refrigerant to the specified amount.

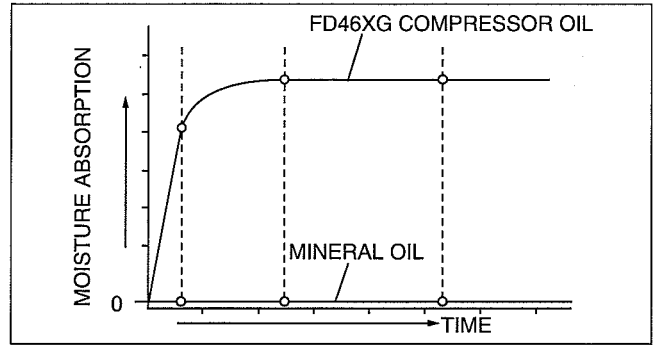
Handling A/C Related Parts

- There will be no effect on either performance or quality of A/C-related parts, including but not limited to the piping, etc., provided that said parts are serviced/repared in accordance with standard procedure.
- When serviced/repared A/C units in locations where there is a considerable risk of contamination from dirt or dust, plug up the openings on the parts to prevent dust/dirt from entering.
- When using adhesive material to cover openings on A/C parts, make sure that the adhesive side of the material does not come in contact with the connectors on the parts.
- When the system is opened to atmosphere, there is no need to replace the condenser, or the receiver dryer when it is housed within the condenser, provided that the unit is plugged accordingly.

REFRIGERANT SYSTEM

Handling Compressor Oil

- Use only FD46XG compressor oil for this vehicle. Using a PAG oil other than FD46XG compressor oil can damage the A/C compressor.
- Do not spill FD46XG compressor oil on the vehicle. A drop of compressor oil on the vehicle surface can eat away at the paint. If oil gets on the vehicle, wipe it off immediately.
- FD46XG compressor oil (PAG oil) has a higher moisture absorption efficiency than the previously used mineral oil. If moisture mixes with the compressor oil, the refrigerant system could be damaged. Therefore, install caps immediately after using the compressor oil or removing refrigerant system parts to prevent moisture absorption.

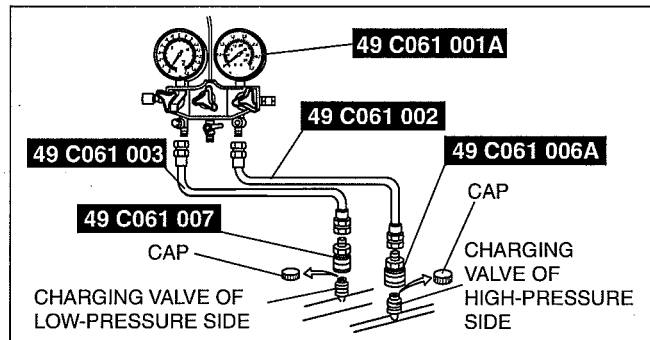


DBG710ZWB001

REFRIGERANT SYSTEM GENERAL PROCEDURES

1. Fully close the valves of the **SST** (49 C061 001A).
2. Connect the **SSTs** (49 C061 002, 49 C061 003) to the high- and low-pressure side joints of the **SST** (49 C061 001A).
3. Connect the **SSTs** (49 C061 006A, 49 C061 007) to the ends of the **SSTs** (49 C061 002, 49 C061 003).
4. Connect the **SSTs** (49 C061 006A, 49 C061 007) to the charging valves.

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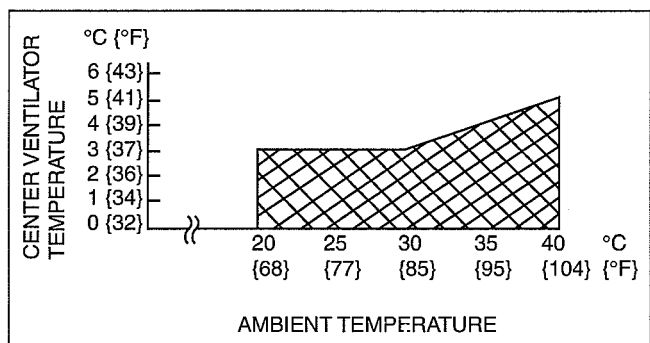
REFRIGERANT SYSTEM PERFORMANCE TEST

1. Inspect the refrigerant pressure. (See 07-10-3 REFRIGERANT PRESSURE CHECK.)
2. Place a dry-bulb thermometer in the driver-side center ventilator outlet.
3. Start the engine and after it is warmed up, run it at a constant **1,500 rpm**.
4. Set the fan speed to MAX HI.
5. Turn the A/C switch on.
6. Set to RECIRCULATE mode.
7. Set the temperature control to MAX COLD.
8. Set to VENT mode.
9. Close all the doors and windows.
10. Wait until the air conditioner output temperature stabilizes.

dcf07100000w04

Stabilized condition

- The A/C compressor repeatedly turns on and off at regular intervals.
11. After the blower air is stabilized, read the dry-bulb thermometer.
 12. Verify the ambient temperature.
 13. Verify that the temperature reading is in the shaded zone.
 - If there is any malfunction, inspect the refrigerant system according to the troubleshooting chart.



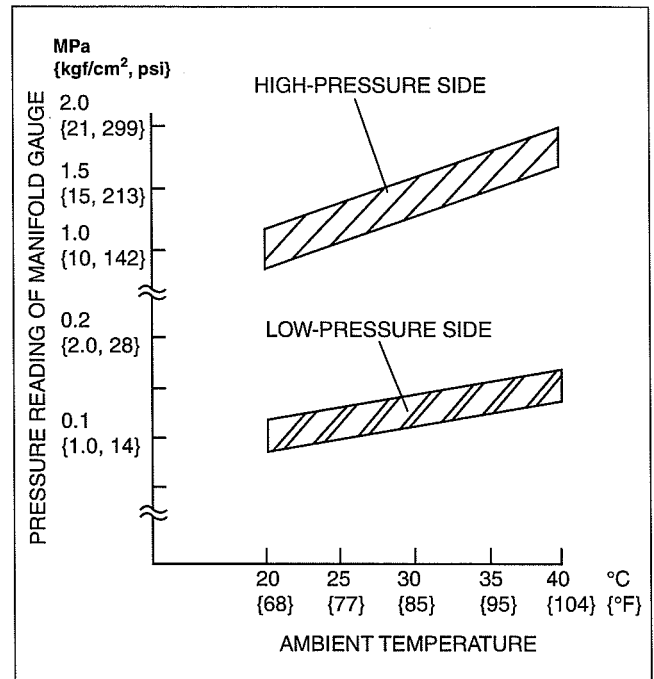
DBG710ZWB003

REFRIGERANT SYSTEM

REFRIGERANT PRESSURE CHECK

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1. Install the **SSTs** (49 C061 0A0B).
2. Start the engine and after it is warmed up, run it at a constant **1,500 rpm**.
3. Set the fan speed **MAX HI**.
4. Turn the **A/C switch** on.
5. Set to **RECIRCULATE** mode.
6. Set the temperature control to **MAX COLD**.
7. Set to **VENT** mode.
8. Close all the doors and all the windows.
9. Measure the ambient temperature and high- and low- pressure side reading of the **SST** (49 C061 001A).
10. Verify that the intersection of the pressure reading of the **SST** (49 C061 001A) and ambient temperature is in the shaded zone.
 - If there is any malfunction, inspect the refrigerant system according to the troubleshooting chart.



DBG710ZW002

RECOVERY

dcf071078834w02

1. Connect an R-134a recovery/recycling/recharging device to the vehicle and follow the device manufacturer's instructions.

REFRIGERANT CHARGING

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Caution

- Do not exceed the specification when charging the system with refrigerant. Doing so will decrease the efficiency of the air conditioner or damage the refrigeration cycle parts.

Charging Recycled R-134a Refrigerant

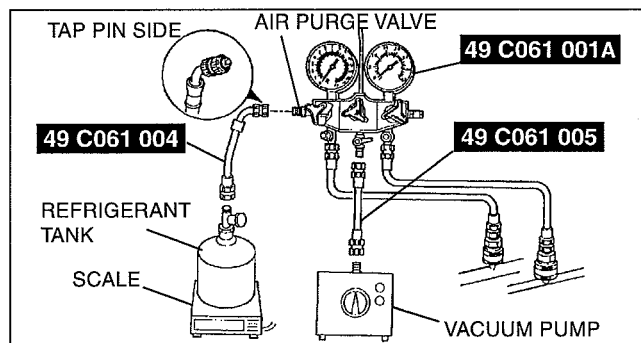
1. Connect an R-134a recovery/recycling/recharging device to the vehicle and follow the device manufacturer's instructions.

REFRIGERANT SYSTEM

Charging Preparation

1. Install the **SSTs** (49 C061 0A0B).
2. Connect the tap pin side of the **SST** (49 C061 004) to the air purge valve of the **SST** (49C061 001).
3. Connect the **SST** (49 C061 005) to the center joint of the **SST** (49 C061 001A).
4. Connect the **SST** (49 C061 005) to the vacuum pump.
5. Connect the **SST** (49 C061 004) to the refrigerant tank.
6. Place the refrigerant tank on the scale.

Regular amount of refrigerant (approx.
quantity)
475 g {16.8 oz}



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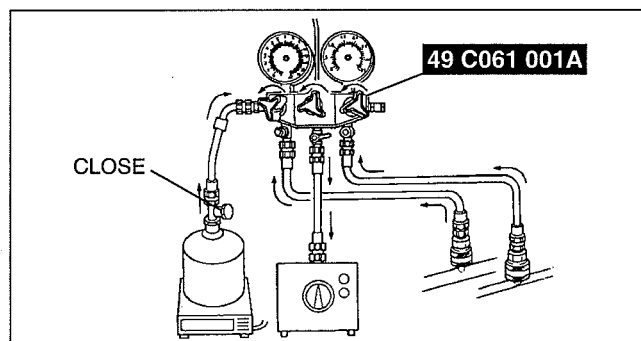
Evacuation

1. Open all the valves of the **SST** (49 C061 001A).

Caution

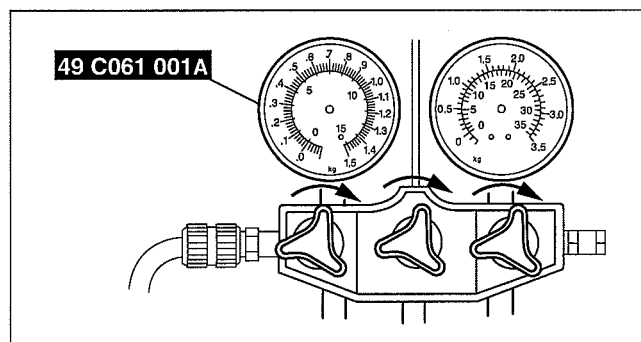
- Close the **SST** (49 C061 001A) valve immediately after stopping the vacuum pump. If the valve is left open, the vacuum pump oil will flow back into the refrigeration cycle and cause a decrease in the efficiency of the air conditioner.

2. Start the vacuum pump and let it operate for 15 min.



BHE0710W005

3. Verify that high- and low-pressure side readings of the **SST** (49 C061 001A) are at **-101 kPa {-760 mmHg, -29.9 inHg}**. Close each valve of the **SST** (49 C061 001A).



BHE0710W002

Airtightness Check

1. Stop the vacuum pump and wait for 5 min.
2. Check the high- and low-pressure side readings of the **SST** (49 C061 001A).
 - If the reading has changed, inspect for leakage and go to Evacuation. (See 07-10-4 Evacuation.)
 - If the reading has not changed, go to Charging New R-134a Refrigerant. (See 07-10-5 Charging New R-134a Refrigerant.)

REFRIGERANT SYSTEM

Charging New R-134a Refrigerant

1. Open the valve of the refrigerant tank.
2. Weigh the refrigerant tank to charge the suitable amount of refrigerant.

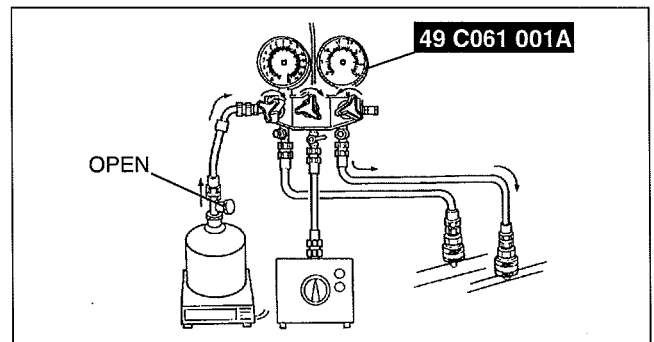
Warning

- If the refrigerant system is charged with a large amount of refrigerant when inspecting for gas leakage, and if any leakage should occur, the refrigerant will be released into the atmosphere. In order to prevent the accidental release of refrigerant which can destroy the ozone layer in the stratosphere, follow the proper procedures and charge with only a small amount of refrigerant when inspecting for gas leakage.
- If charging the system with refrigerant using service cans, running the engine with the high-pressure side valve open is dangerous. Pressure within the service cans will increase and the cans could explode, scattering metal fragments and liquid refrigerant that can seriously injure you. Therefore, do not open the high-pressure side valve while the engine is running.

Caution

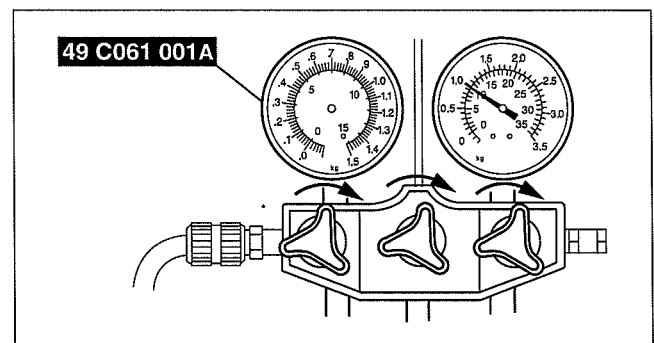
- Always being charging of refrigerant from the high-pressure side. If changing is begun from the low-pressure side, the vanes of the A/C compressor will not be released and abnormal noise may result.

3. Open the low-pressure side valve of the SST (49 C061 001A).



BHE0710W006

4. When the high-pressure side reading increases to **0.098 MPa {1.0 kgf/cm², 14 psi}**, close the low-pressure side valve of the SST (49 C061 001A).
5. Inspect for leakage from the cooler pipe/hose connections using the SST (49 C061 013).
 - If there is no leakage, go to Step 7.
 - If leakage is found at a loose joint, tighten the joint, then go to next step.
6. Inspect for leakage again.
 - If there is no leakage after tightening the joint, go to next step.
 - If there is still a leakage at the same joint, discharge the refrigerant and then repair the joint. Repeat the charging procedure from evacuation.



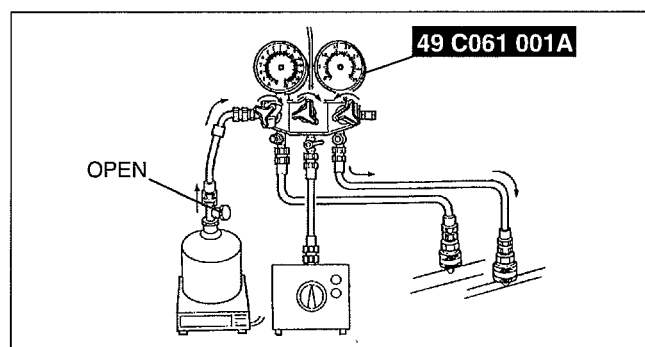
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Warning

- If charging the system with refrigerant using service cans, running the engine with the high-pressure side valve open is dangerous. Pressure within the service cans will increase and the cans could explode, scattering metal fragments and liquid refrigerant that can seriously injure you. Therefore, do not open the high-pressure side valve while the engine is running.

REFRIGERANT SYSTEM

7. Open the low-pressure side valve of the **SST** (49 C061 001A) and charge with refrigerant until the weight of refrigerant tank has decreased **250 g {8.83 oz}** from the amount in Step 2.

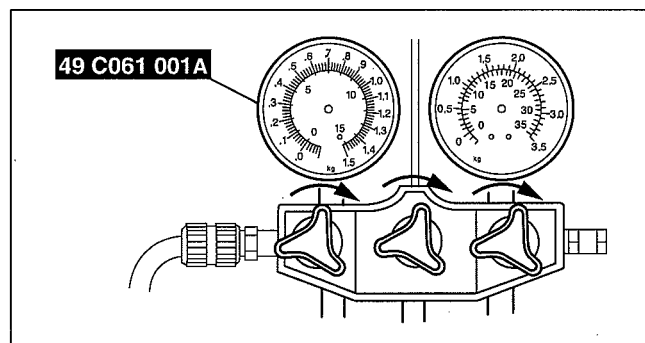


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8. Close the low-pressure side valve of the **SST** (49 C061 001A).

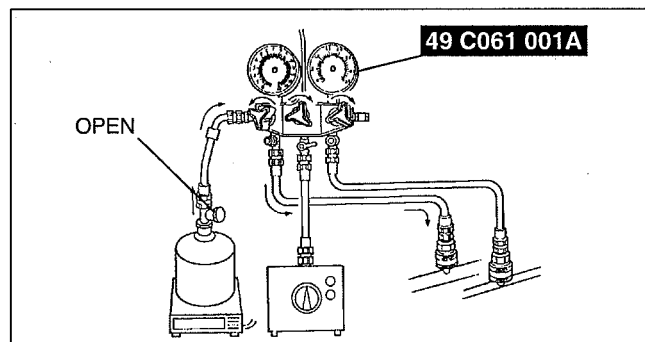
Warning

- If charging the system with refrigerant using service cans, running the engine with the high-pressure side valve open is dangerous. Pressure within the service cans will increase and the cans could explode, scattering metal fragments and liquid refrigerant that can seriously injure you. Therefore, do not open the high-pressure side valve while the engine is running.



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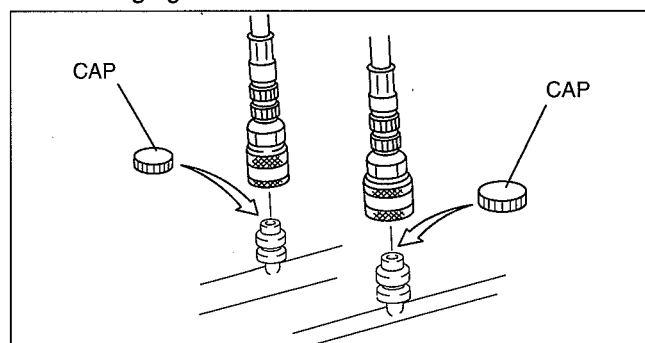
9. Start the engine and actuate the A/C compressor.
10. Open the low-pressure side valve of the **SST** (49 C061 001A) and charge with refrigerant until the weight of the refrigerant tank has decreased regular amount from the amount in Step 2.
11. Close the low-pressure side valve of the **SST** (49 C061 001A) and the valve of the refrigerant tank.
12. Stop the engine and A/C compressor.



BHE0710W007

Leak Test

1. Inspect for leakage using the **SST** (49 C061 013).
 - If there is no leakage, go to Step 3.
 - If leakage is found at a loose joint, tighten the joint, then go to next step.
2. Inspect for leakage again.
 - If there is no leakage after tightening the joint, go to next step.
 - If there is still leakage at the same joint, discharge the refrigerant and then repair the joint. Repeat the charging procedure from evacuation.
3. Disconnect the **SSTs** (49 C061 006A, 49 C061 007) from the charging valves.
4. Install the caps to the charging valves.



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BASIC SYSTEM

07-11 BASIC SYSTEM

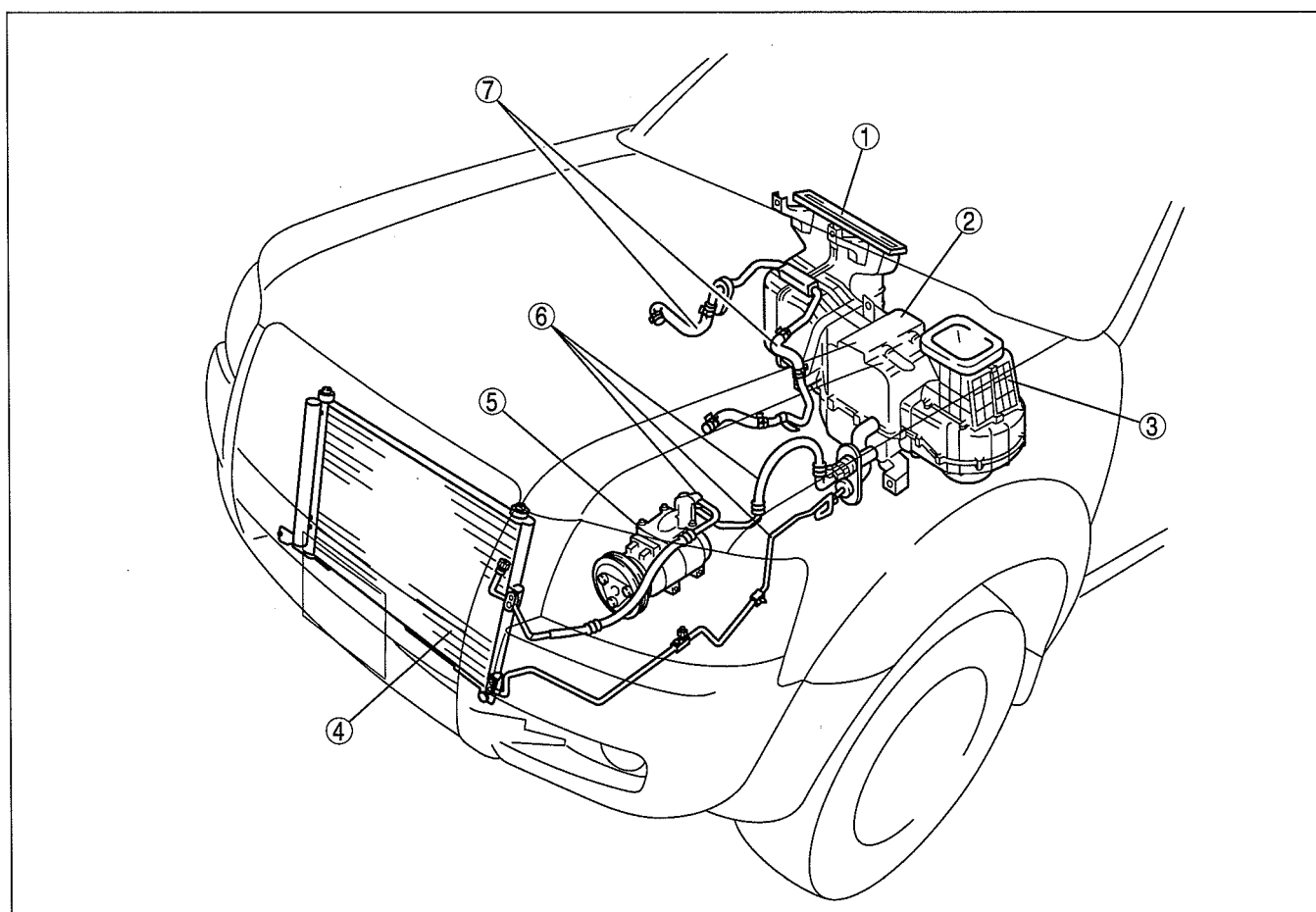
HVAC BASIC SYSTEM

LOCATION INDEX.....	07-11-1
BLOWER UNIT	
REMOVAL/INSTALLATION.....	07-11-2
BLOWER UNIT	
DISASSEMBLY/ASSEMBLY.....	07-11-2
COOLING UNIT	
REMOVAL/INSTALLATION.....	07-11-3
COOLING UNIT	
DISASSEMBLY/ASSEMBLY.....	07-11-4
EVAPORATOR INSPECTION.....	07-11-5

AIR OUTLET DUCT

REMOVAL/INSTALLATION.....	07-11-5
AIR OUTLET DUCT	
DISASSEMBLY/ASSEMBLY.....	07-11-6
A/C COMPRESSOR	
REMOVAL/INSTALLATION.....	07-11-6
CONDENSER	
REMOVAL/INSTALLATION.....	07-11-8
CONDENSER INSPECTION.....	07-11-8
REFRIGERANT LINES	
REMOVAL/INSTALLATION.....	07-11-8

HVAC BASIC SYSTEM LOCATION INDEX



arnffw0000449

1	Heater unit
2	Cooling unit (See 07-11-2 COOLING UNIT REMOVAL/ INSTALLATION.) (See 07-11-4 COOLING UNIT DISASSEMBLY/ ASSEMBLY.)
3	Blower unit (See 07-11-5 BLOWER UNIT DISASSEMBLY/ ASSEMBLY.)
4	Condenser

5	A/C compressor (See 07-11-6 A/C COMPRESSOR REMOVAL/ INSTALLATION.)
6	Refrigerant line (See 07-11-8 REFRIGERANT LINES REMOVAL/ INSTALLATION.)
7	Heater hose

BASIC SYSTEM

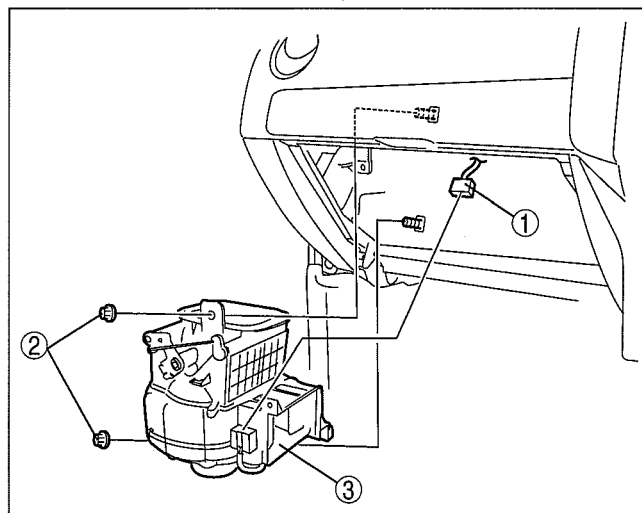
BLOWER UNIT REMOVAL/INSTALLATION

dcf071161140w01

1. Disconnect the negative battery cable.
2. Remove the glove compartment. (See 09-17-9 GLOVE COMPARTMENT REMOVAL/INSTALLATION.)
3. Remove the cooling unit. (See 07-11-3 COOLING UNIT REMOVAL/INSTALLATION.)
4. Remove in the order indicated in the table.

1	Connector
2	Nut
3	Blower unit

5. Install in the reverse order of removal.



DBG711ZWB002

BLOWER UNIT DISASSEMBLY/ASSEMBLY

dcf071161140w02

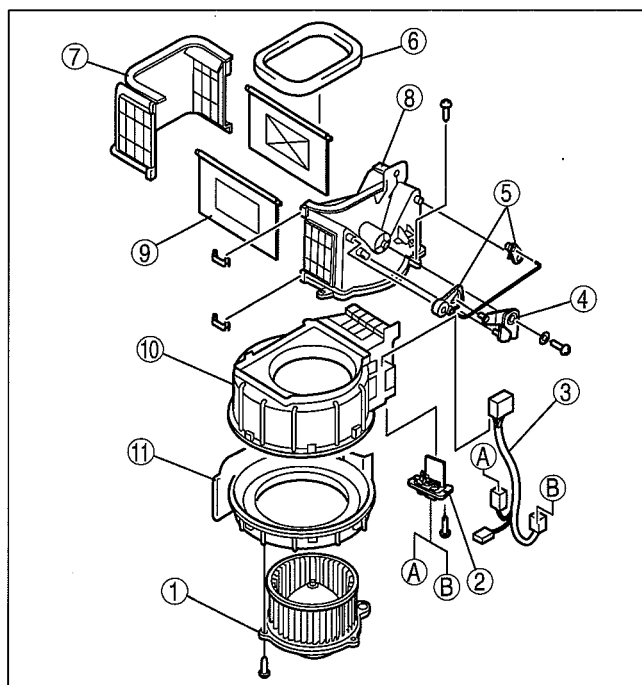
1. Disassemble in the order indicated in the table.

Caution

- Apply only the specified grease to the link. Otherwise abnormal noise or improper operation may result.

1	Blower motor
2	Resistor
3	Blower wiring harness
4	Air intake link
5	Air intake crank
6	Polyurethane protector
7	Blower case (1)
8	Blower case (2)
9	Air interke door
10	Blower case (3)
11	Blower case (4)

2. Assemble in the reverse order of disassembly.



DBG711ZWB027

BASIC SYSTEM

COOLING UNIT REMOVAL/INSTALLATION

dcf071161520w01

1. Disconnect the negative battery cable.
2. Discharge the refrigerant from the system. (See 07-10-3 REFRIGERANT CHARGING.)
3. Remove the glove compartment. (See 09-17-9 GLOVE COMPARTMENT REMOVAL/INSTALLATION.)
4. Disconnect the air intake wire from the blower unit. (See 07-40-8 CLIMATE CONTROL UNIT WIRE ADJUSTMENT.)

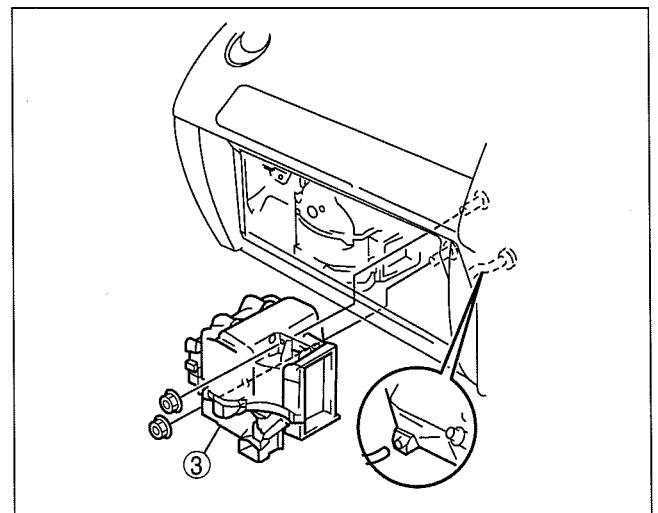
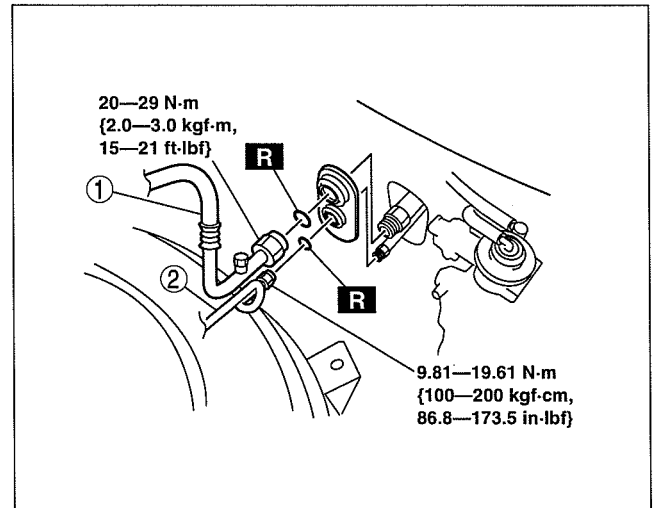
Caution

- If moisture or foreign material enters the refrigeration cycle, cooling ability will be lowered and abnormal noise will occur. Always immediately plug all open fittings after removing any refrigeration cycle parts to keep moisture or foreign material out of the cycle.

5. Remove in the order indicated in the table. Do not allow compressor oil to spill.

1	Cooler hose (LO) (See 07-11-9 Refrigerant Line Removal Note.) (See 07-11-9 Refrigerant Line Installation Note.)
2	Cooler pipe (See 07-11-9 Refrigerant Line Removal Note.) (See 07-11-9 Refrigerant Line Installation Note.)
3	Cooling unit (See 07-11-3 Cooling Unit Installation Note.)

6. Install in the reverse order of removal.
7. Perform the refrigerant system performance test.
(See 07-10-2 REFRIGERANT SYSTEM PERFORMANCE TEST.)



07

Cooling Unit Installation Note

1. When installing a new cooling unit (evaporator), add FD46XG compressor oil into the refrigerant cycle.

Supplemental amount (approx. quantity)
40 ml {40 cc, 1.4 fl oz}

BASIC SYSTEM

COOLING UNIT DISASSEMBLY/ASSEMBLY

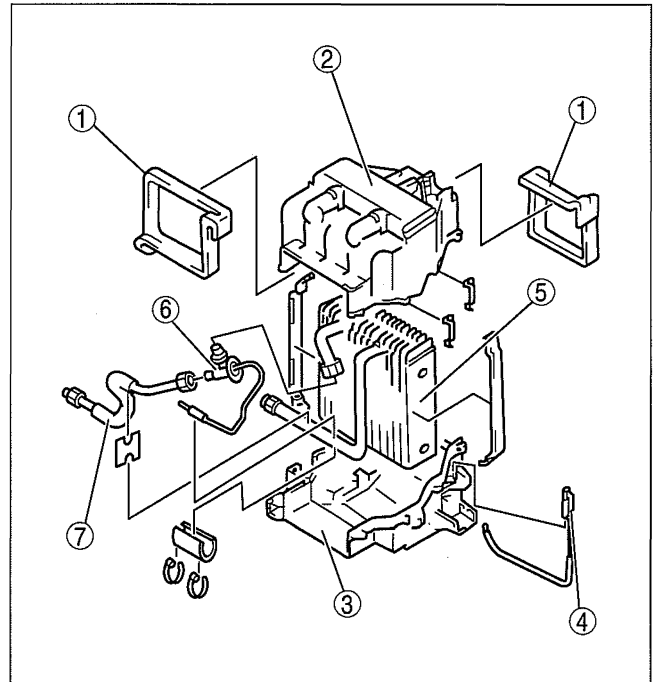
dcf071161520w02

Caution

- If moisture or foreign material enters the refrigeration cycle, cooling ability will be lowered and abnormal noise will occur. Always immediately plug all open fittings after removing any refrigeration cycle parts to keep moisture or foreign material out of the cycle.

1. Disassemble in the order indicated in the table.
2. Assemble in the reverse order of disassembly.

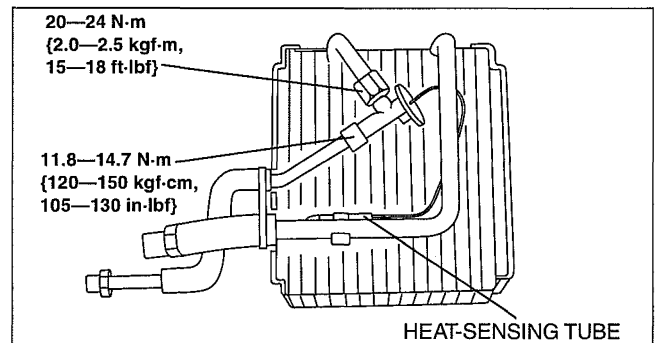
1	Polyurethane protector
2	Cooler case (1)
3	Cooler case (2)
4	Thermistor (See 07-11-5 Thermistor Assembly Note.)
5	Evaporator
6	Expansion valve (See 07-11-4 Expansion Valve assembly Note.)
7	High pressure pipe



DBG711ZWB020

Expansion Valve assembly Note

- Replace the O-rings at the expansion valve joints and apply compressor oil to the O-rings and connect the joints.
- Tighten the joint to the specified torque using two open-end wrenches.
- Assemble the heat-sensing tube as shown in the figure.

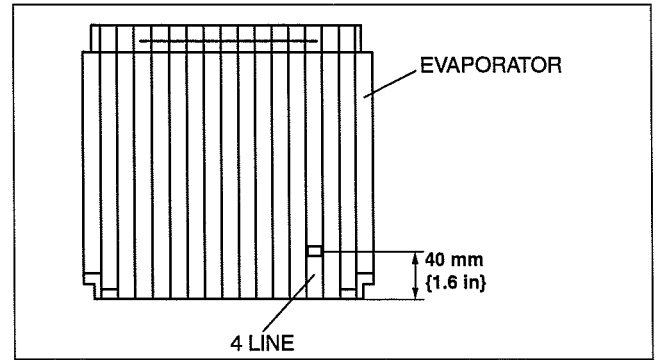


DBG711ZWB009

BASIC SYSTEM

Thermistor Assembly Note

1. Assemble the thermistor as shown in the figure.



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EVAPORATOR INSPECTION

dcf071161810w01

1. Remove the cooling unit. (See 07-11-3 COOLING UNIT REMOVAL/INSTALLATION.)
2. Remove the evaporator from the cooling unit.
3. Check for cracks, damage, and oil leakage. If any problems found, replace the evaporator.
4. Check for bent fins. If they are bent, use a flat-head screwdriver to straighten them.

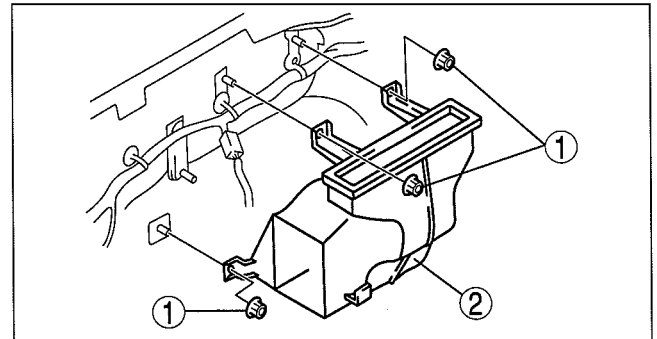
AIR OUTLET DUCT REMOVAL/INSTALLATION

dcf071160120w01

1. Disconnect the negative battery cable.
2. Remove the dashboard. (See 09-17-4 DASHBOARD REMOVAL/INSTALLATION.)
3. Remove as indicated in the table.

1	Nut
2	Air outlet duct

4. Install in the reverse order of removal.



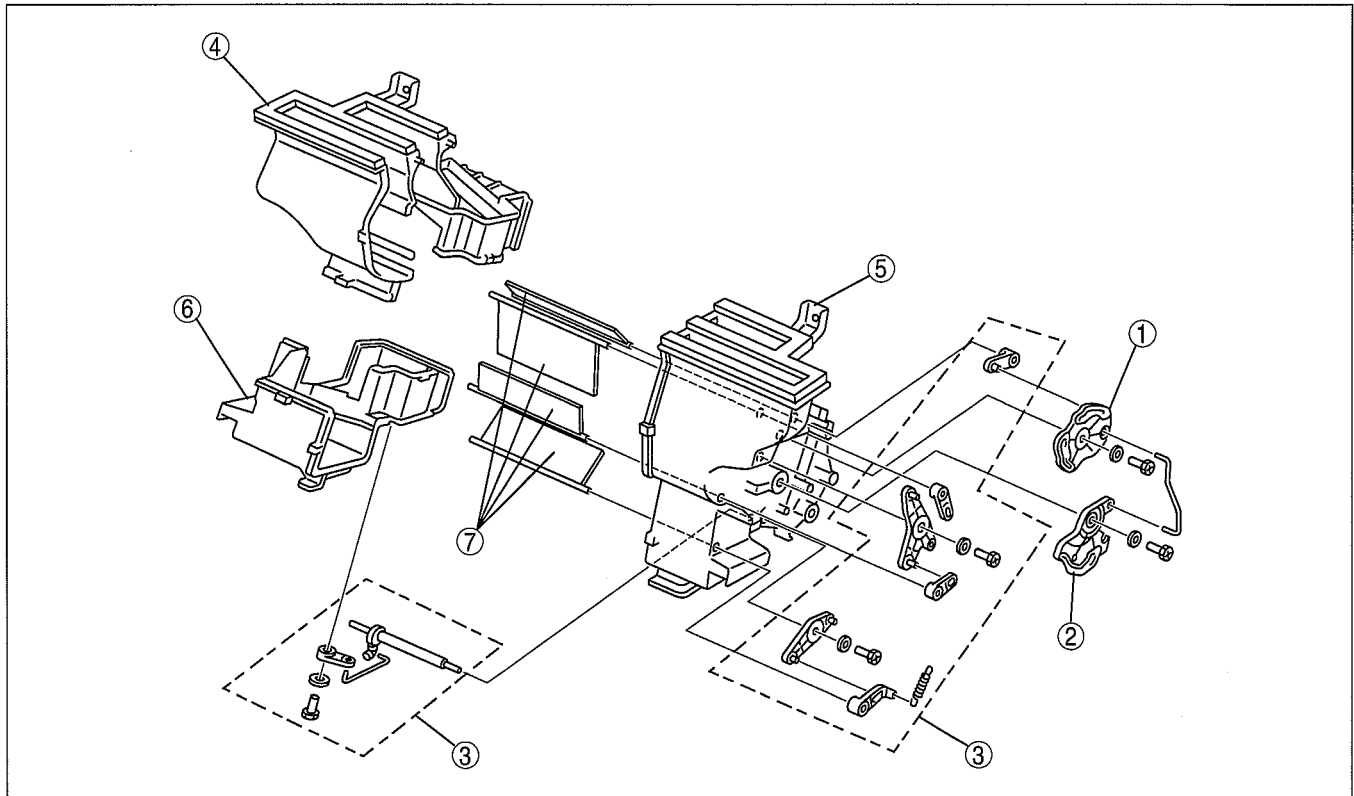
DBG711ZW012

BASIC SYSTEM

AIR OUTLET DUCT DISASSEMBLY/ASSEMBLY

dcf071160120w02

1. Disassemble in the order indicated in the table.



DBG740ZWB033

1	Airflow mode main link
2	Airflow mode link
3	Airflow mode link set
4	Case (1)

5	Case (2)
6	Case (3)
7	Airflow mode door

2. Assemble in the reverse order of disassembly.

A/C COMPRESSOR REMOVAL/INSTALLATION

dcf071161450w01

1. Disconnect the negative battery cable.
2. Discharge the refrigerant from the system. (See 07-10-3 REFRIGERANT CHARGING.)
3. Remove the drive belt.
4. Loosen the A/C compressor bolts, then slide the A/C compressor.

Caution

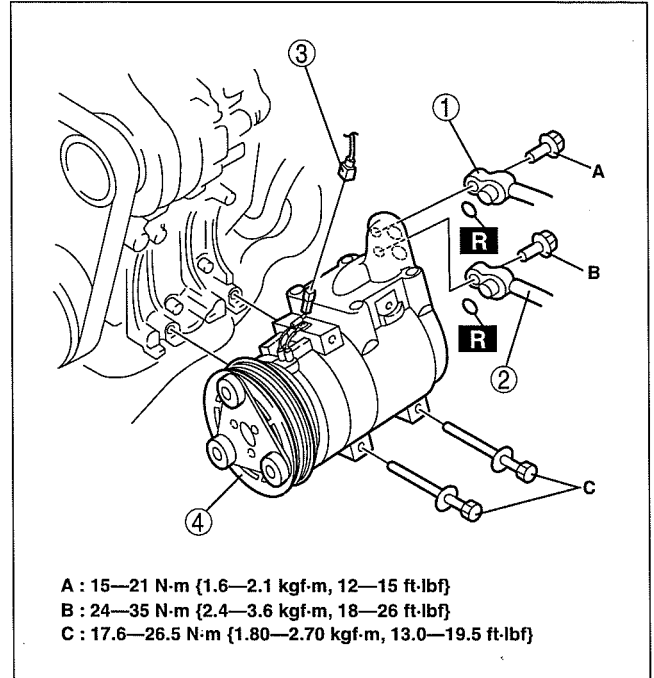
- If moisture or foreign material enters the refrigeration cycle, cooling ability will be lowered and abnormal noise will occur. Always immediately plug all open fittings after removing any refrigeration cycle parts to keep moisture or foreign material out of the cycle.

BASIC SYSTEM

5. Remove in the order indicated in the table. Do not allow compressor oil to spill.

1	Cooler hose (HI)
2	Cooler hose (LO)
3	Magnetic connector
4	A/C compressor (See 07-11-7 A/C Compressor Installation Note.)

6. Install in the reverse order of removal.
7. Adjust the drive belt. (See 01-10A-4 DRIVE BELT ADJUSTMENT [WL-3].) (See 01-10B-4 DRIVE BELT ADJUSTMENT [WL-C, WE-C].)
8. Perform the refrigerant system performance test. (See 07-10-2 REFRIGERANT SYSTEM PERFORMANCE TEST.)



DBG711ZWB024

A/C Compressor Installation Note

Caution

- Due to the high moisture-absorption characteristics of the compressor oil, it may absorb moisture if left over a long period of time thereby negatively affecting A/C operation. Drain the compressor oil and refill within 10 min. of each other.

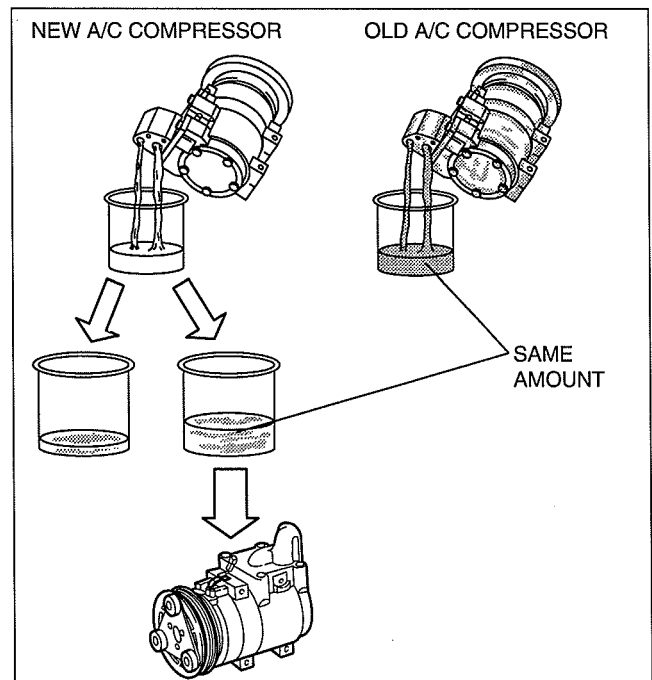
1. Rotate new A/C compressor shaft six to eight revolutions while collecting refrigerant oil in a clean measuring device. Use this refrigerant oil to refill new compressor. Do not allow refrigerant oil to become contaminated.
2. Rotate old A/C compressor shaft six to eight revolutions while collecting refrigerant oil in a separate, clean measuring device.
3. Compare those oil amounts. The amount of the oil drained from the new A/C compressor should be greater than the old one.
4. Pour the same amount oil of drained from the old A/C compressor back into the new A/C compressor.

A/C compressor oil type

- FD46XG

A/C compressor oil sealed volume (approx. quantity)

- 180 ml {180 cc, 6.08 fl oz}



DBG711ZWB016

BASIC SYSTEM

CONDENSER REMOVAL/INSTALLATION

dcf071161480w01

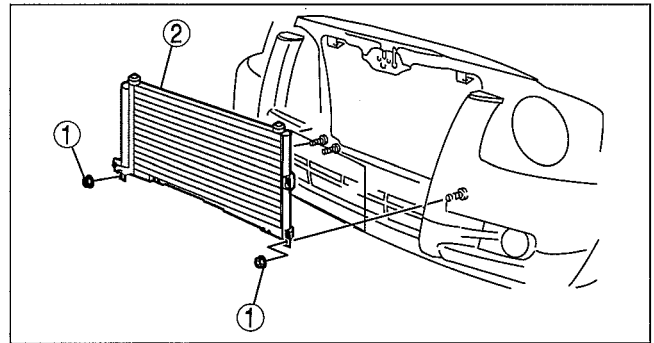
1. Disconnect the negative battery cable.
2. Discharge the refrigerant from the system. (See 07-10-3 REFRIGERANT CHARGING.)
3. Remove the radiator grill. (See 09-16-3 RADIATOR GRILLE REMOVAL/INSTALLATION.)
4. Remove the charge air cooler. (See 01-13A-2 INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [WL-3].) (See 01-13B-3 INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [WL-C, WE-C].)
5. Disconnect the cooler hose (HI) and cooler pipe. Do not allow remaining compressor oil in the refrigerant line to spill.
6. Remove in the order indicated in the table.

Caution

- If moisture or foreign material enters the refrigeration cycle, cooling ability will be lowered and abnormal noise or other malfunction could occur. Always plug open fittings immediately after removing any refrigeration cycle parts.

1	Nut
2	Condenser (See 07-11-8 Condenser Installation Note.)

7. Install in the reverse order of removal.
8. Perform the refrigerant system performance test. (See 07-10-2 REFRIGERANT SYSTEM PERFORMANCE TEST.)



DBG711ZWB029

Condenser Installation Note

1. After replacing the condenser, add compressor oil to the refrigeration cycle.

Supplemental oil amount (approx. quantity)
20 ml {20 cc, 0.7 fl oz}

CONDENSER INSPECTION

dcf071161480w02

1. Inspect the condenser for cracks, damage, and oil leakage.
 - If there is any malfunction, replace the condenser.
2. Visually inspect the fins for clogging of foreign material.
 - If any fins are clogged, remove the foreign material.
3. Visually inspect the fins for bending.
 - If there is any bending, use the end of a flathead screwdriver to straighten fins.

REFRIGERANT LINES REMOVAL/INSTALLATION

dcf071161460w01

1. Disconnect the negative battery cable.
2. Discharge the refrigerant from the system. (See 07-10-3 RECOVERY.) (See 07-10-3 REFRIGERANT CHARGING.)

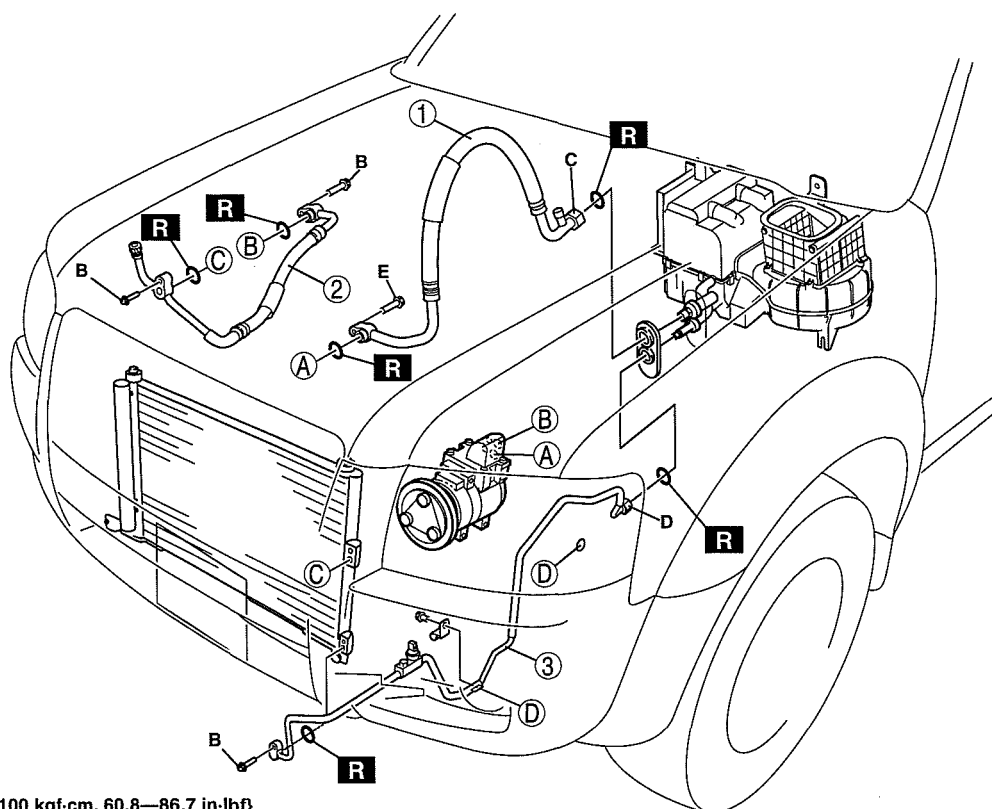
Caution

- If moisture or foreign material enters the refrigeration cycle, cooling ability will be lowered and abnormal noise will occur. Always immediately plug all open fittings after removing any refrigeration cycle parts to keep moisture or foreign material out of the cycle.

3. Remove the radiator grille. (See 09-16-3 RADIATOR GRILLE REMOVAL/INSTALLATION.)

BASIC SYSTEM

4. Remove in the order indicated in the table. Do not allow compressor oil to spill.



- A: 6.85—9.80 N·m {70—100 kgf·cm, 60.8—86.7 in·lb}
 B: 15—21 N·m {1.5—2.2 kgf·m, 11—15 ft·lb}
 C: 20—29 N·m {2.0—3.0 kgf·m, 15—21 ft·lb}
 D: 9.81—19.61 N·m {100—200 kgf·cm, 86.8—173.5 in·lb}
 E: 24—35 N·m {2.4—3.6 kgf·m, 18—26 ft·lb}

DBG711ZW028

1	Cooler hose (LO) (See 07-11-9 Cooler Hose Removal Note.) (See 07-11-9 Refrigerant Line Removal Note.) (See 07-11-9 Refrigerant Line Installation Note.)
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2	Cooler hose (HI) (See 07-11-9 Cooler Hose Removal Note.) (See 07-11-9 Refrigerant Line Removal Note.) (See 07-11-9 Refrigerant Line Installation Note.)
3	Cooler pipe (See 07-11-9 Refrigerant Line Removal Note.) (See 07-11-9 Refrigerant Line Installation Note.)

5. Install in the reverse order of removal.

6. Perform the refrigerant system performance test. (See 07-10-2 REFRIGERANT SYSTEM PERFORMANCE TEST.)

Cooler Hose Removal Note

WL-3 only

- Loosen the A/C compressor bolts, then slide the A/C compressor, and remove the cooler hose bolt from the A/C compressor.

Refrigerant Line Removal Note

- Loosen the nut by using two spanners, then remove the cooler pipe or hose.

Refrigerant Line Installation Note

- When installing a new cooler pipe or hose, add FD46XG compressor oil into the refrigeration cycle.

Supplemental amount

- Cooler hose (LO): 8 ml {8 cc, 0.3 fl oz}
- Cooler hose (HI): 5 ml {5 cc, 0.2 fl oz}
- Cooler pipe: 3 ml {3 cc, 0.1 fl oz}

2. Apply compressor oil to the O-rings and connect the joints.

3. Tighten the joints.

(1) Tighten the nut or bolt of the joint by hand.

(2) Tighten the joint to the specified torque. If it is a nut joint, tighten the nut using a spanner and torque wrench.

CONTROL SYSTEM

07-40 CONTROL SYSTEM

HVAC CONTROL SYSTEM LOCATION

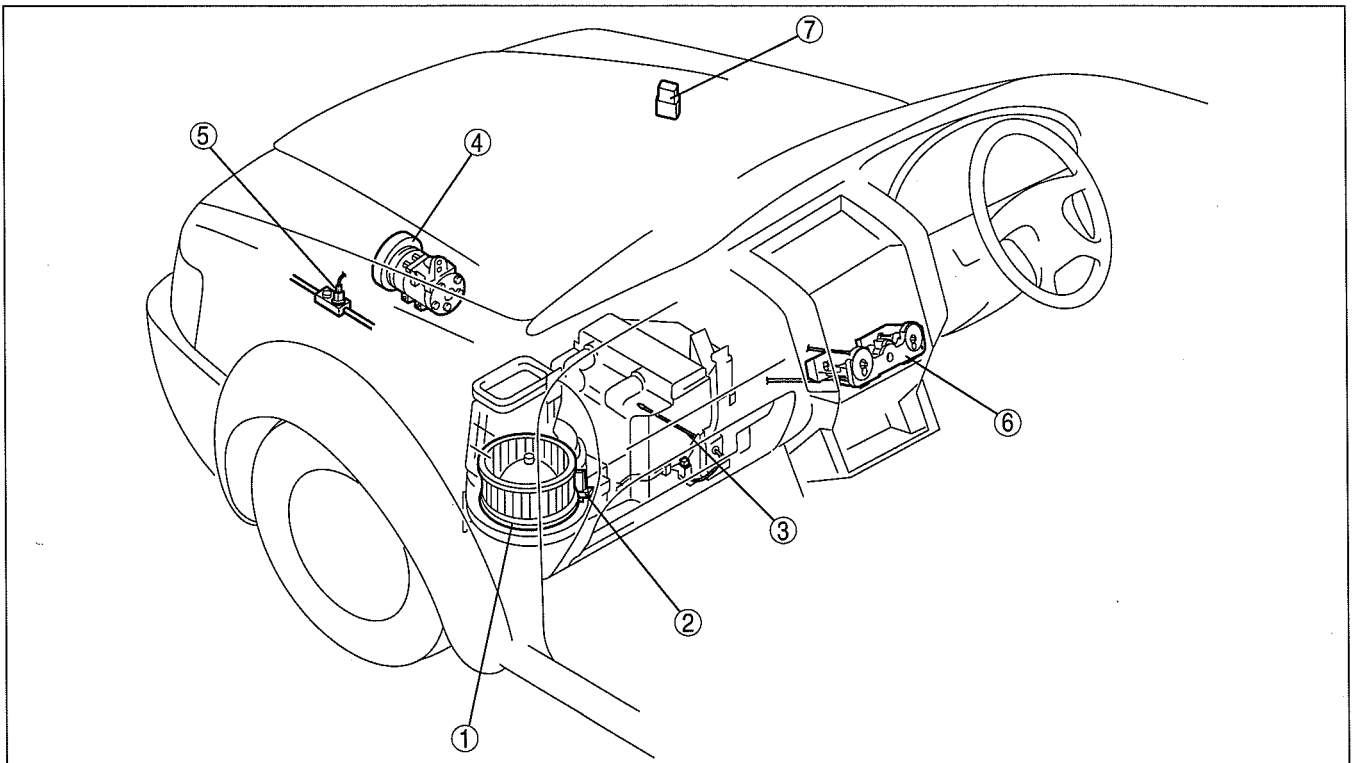
INDEX	07-40-1
BLOWER MOTOR	
REMOVAL/INSTALLATION.....	07-40-2
BLOWER MOTOR INSPECTION	07-40-2
RESISTOR	
REMOVAL/INSTALLATION.....	07-40-3
RESISTOR INSPECTION	07-40-3
MAGNETIC CLUTCH	
DISASSEMBLY/ASSEMBLY.....	07-40-3
MAGNETIC CLUTCH ADJUSTMENT...	07-40-4
MAGNETIC CLUTCH INSPECTION....	07-40-5
THERMISTOR	
REMOVAL/INSTALLATION.....	07-40-5
THERMISTOR INSPECTION	07-40-5

REFRIGERANT PRESSURE SWITCH

REMOVAL/INSTALLATION	07-40-6
REFRIGERANT PRESSURE SWITCH	
INSPECTION	07-40-6
CLIMATE CONTROL UNIT	
REMOVAL.....	07-40-7
CLIMATE CONTROL UNIT	
INSTALLATION.....	07-40-7
CLIMATE CONTROL UNIT	
DISASSEMBLY/ASSEMBLY	07-40-8
CLIMATE CONTROL UNIT WIRE	
ADJUSTMENT	07-40-8
A/C AMPLIFIER INSPECTION.....	07-40-9
A/C SWITCH	
REMOVAL/INSTALLATION	07-40-10
A/C SWITCH INSPECTION	07-40-10
FAN SWITCH INSPECTION.....	07-40-10

HVAC CONTROL SYSTEM LOCATION INDEX

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DBG740ZWB01

1	Blower motor (See 07-40-2 BLOWER MOTOR REMOVAL/ INSTALLATION.) (See 07-40-2 BLOWER MOTOR INSPECTION.)
2	Resistor (See 07-40-3 RESISTOR REMOVAL/ INSTALLATION.) (See 07-40-3 RESISTOR INSPECTION.)
3	Thermistor (See 07-40-5 THERMISTOR REMOVAL/ INSTALLATION.) (See 07-40-5 THERMISTOR INSPECTION.)

4	Magnetic clutch (See 07-40-3 MAGNETIC CLUTCH DISASSEMBLY/ASSEMBLY.) (See 07-40-4 MAGNETIC CLUTCH ADJUSTMENT.) (See 07-40-5 MAGNETIC CLUTCH INSPECTION.)
5	Refrigerant pressure switch (See 07-40-6 REFRIGERANT PRESSURE SWITCH REMOVAL/INSTALLATION.) (See 07-40-6 REFRIGERANT PRESSURE SWITCH INSPECTION.)

CONTROL SYSTEM

6	Climate control unit (See 07-40-7 CLIMATE CONTROL UNIT REMOVAL.) (See 07-40-7 CLIMATE CONTROL UNIT INSTALLATION.) (See 07-40-8 CLIMATE CONTROL UNIT DISASSEMBLY/ASSEMBLY.) (See 07-40-8 CLIMATE CONTROL UNIT WIRE ADJUSTMENT.)
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7	A/C relay
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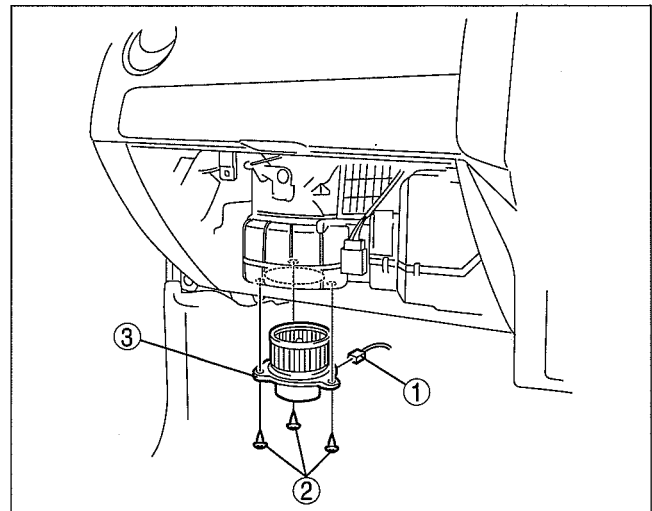
BLOWER MOTOR REMOVAL/INSTALLATION

1. Disconnect negative battery cable.
2. Remove in the order indicated in the table.

1	Blower motor connector
2	Screw
3	Blower motor

3. Install in the reverse order of removal.

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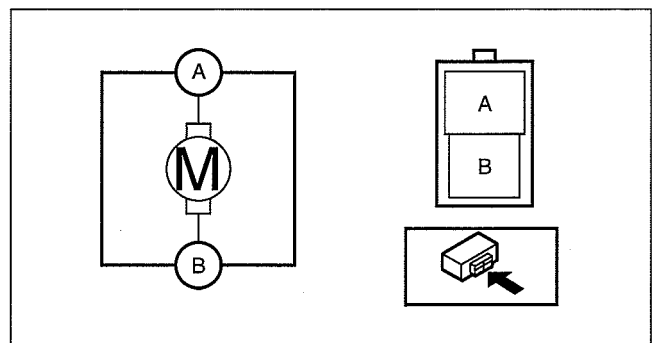
DBG740ZWB016

BLOWER MOTOR INSPECTION

Connect battery positive voltage to blower motor terminal A, connect terminal B to ground, and then verify its operation.

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- If there is any malfunction, replace the blower motor.



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CONTROL SYSTEM

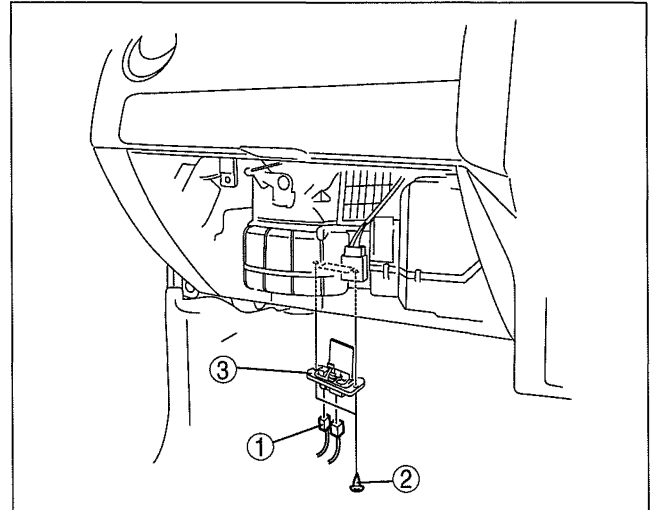
RESISTOR REMOVAL/INSTALLATION

dcf074061015w01

1. Disconnect the negative battery cable.
2. Remove in the order indicated in the table.

1	Resistor connector
2	Screw
3	Resistor

3. Install in the reverse order of removal.



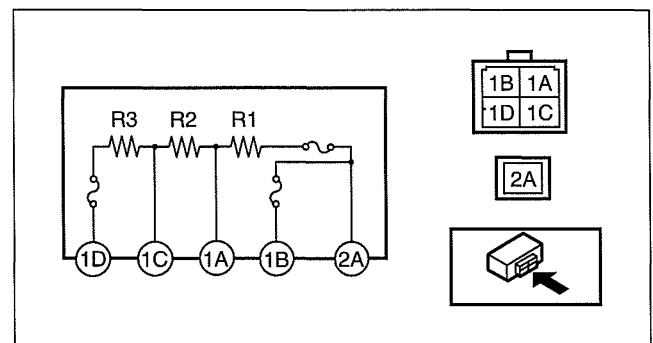
DBG740ZWB018

RESISTOR INSPECTION

dcf074061015w02

1. Verify that the resistance between the resistor terminals is as shown in the table.
 - If there is any malfunction, replace the resistor.

Terminal	Resistance (ohm)
1A—1B (2A)	R1: 0.43—0.47
1A—1C	R2: 1.04—1.16
1C—1D	R3: 1.47—1.63



DBG740ZWB019

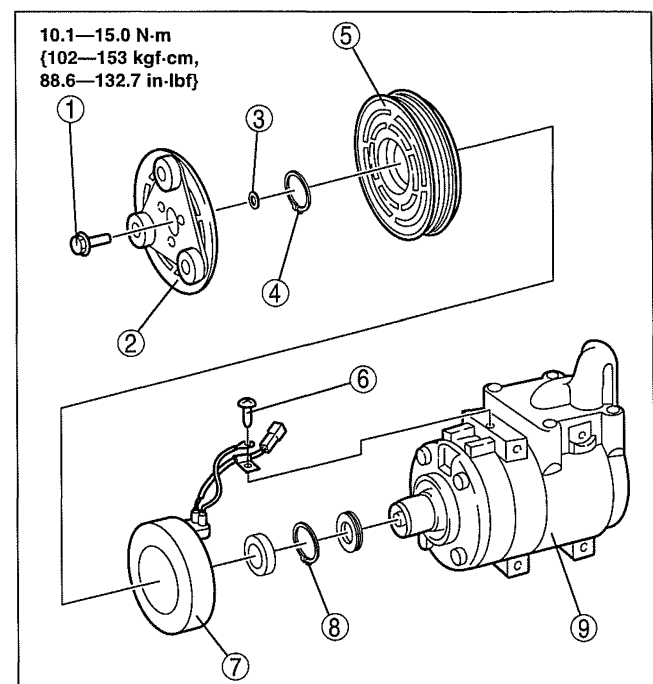
MAGNETIC CLUTCH DISASSEMBLY/ASSEMBLY

dcf074061010w01

1. Disassemble in the order indicated in the table.

1	Bolt (See 07-40-4 Bolt Removal/Installation Note.)
2	Pressure plate (See 07-40-4 Pressure Plate Installation Note.)
3	Shim
4	Snap ring
5	A/C compressor pulley
6	Screw
7	Stator (See 07-40-4 Stator Removal Note.) (See 07-40-4 Stator Installation Note.)
8	Snap ring
9	A/C compressor body

2. Assemble in the reverse order of disassembly.
3. Adjust the magnetic clutch clearance. (See 07-40-4 MAGNETIC CLUTCH ADJUSTMENT.)

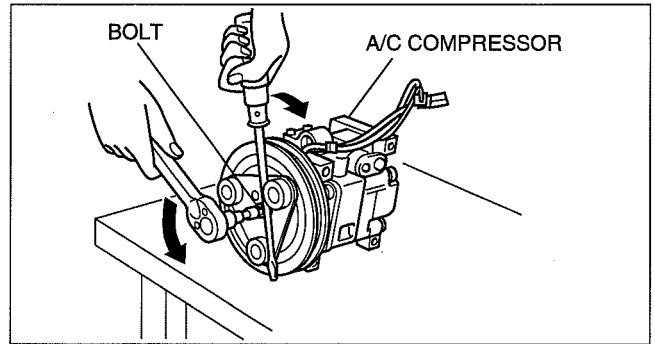


DBG740ZWB002

CONTROL SYSTEM

Bolt Removal/Installation Note

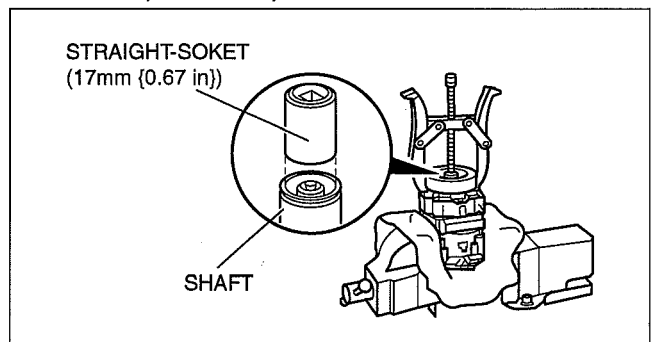
1. When removing or installing the bolt, hold the pressure plate in place as shown in the figure.
2. When installing a new A/C compressor body, replace the recommended bolt.



DBG740ZWB006

Stator Removal Note

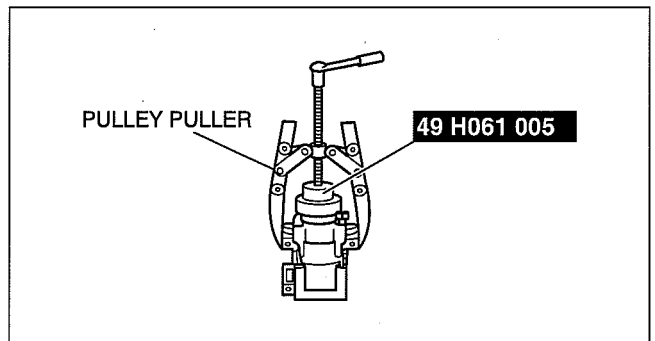
1. Set a straight-socket (17 mm {0.67 in}) on the shaft of the A/C compressor to protect it.
2. Remove the stator using a pulley puller as shown in the figure.



DBG740ZWB028

Stator Installation Note

1. Install the stator perpendicularly using the SST (49 H061 005) and a pulley puller as shown in the figure.



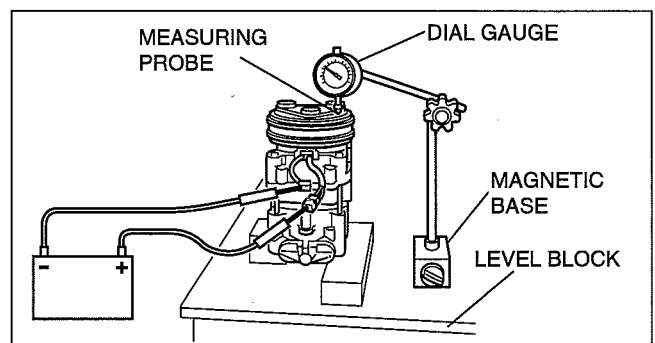
DBG740ZWB029

Pressure Plate Installation Note

1. When installing the pressure plate, carry out magnetic clutch clearance adjustment.

MAGNETIC CLUTCH ADJUSTMENT

1. Set the A/C compressor on a level block.
2. Turn on the magnetic clutch by connecting the battery voltage to the magnetic clutch connector terminal and the ground to the A/C compressor body.
3. Fix a dial gauge on a magnetic base and set the measuring probe onto point A on the pressure plate surface.
4. Turn off the magnetic clutch by disconnecting the ground from the A/C compressor body, then measure the dial gauge readings.



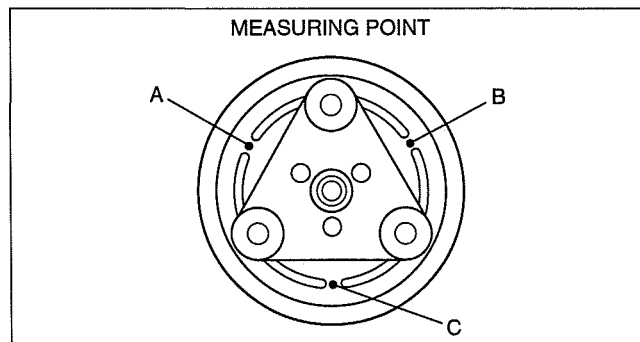
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CONTROL SYSTEM

5. Measure the clearance for point B and C on the pressure plate surface by repeating the above steps 2 through 4.

Clearance

0.35—0.75 mm {0.014—0.029 in}

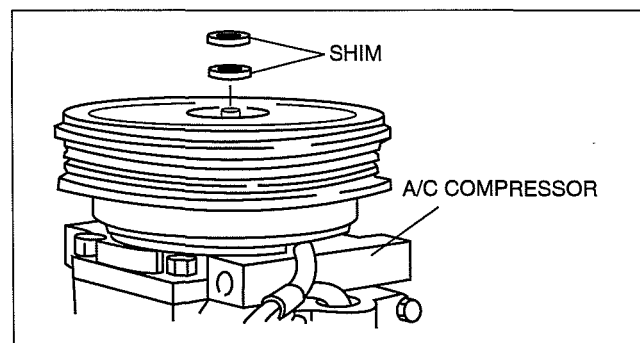


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6. If the clearance is not as specified, adjust it by changing the shim.

Note

- The seven shim sizes from 0.7 mm {0.028 in} to 1.3 mm {0.051 in} differ in 0.1 mm {0.004 in} increments.

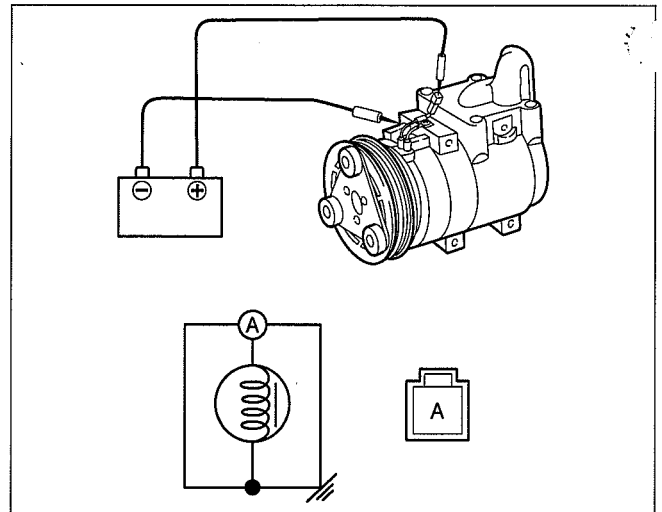


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CONTROL SYSTEM

MAGNETIC CLUTCH INSPECTION

1. Connect battery positive voltage to terminal A of magnetic clutch and the ground to A/C compressor body.
2. Verify that the magnetic clutch operates.
 - If there is any malfunction, replace the stator.



THERMISTOR REMOVAL/INSTALLATION

1. Remove the thermistor from the cooling unit. (See 07-11-4 COOLING UNIT DISASSEMBLY/ASSEMBLY.)

THERMISTOR INSPECTION

1. Disconnect the negative battery cable.
2. Remove the cooling unit. (See 07-11-3 COOLING UNIT REMOVAL/INSTALLATION.)
3. Remove the thermistor. (See 07-11-4 COOLING UNIT DISASSEMBLY/ASSEMBLY.)
4. Connect the negative battery cable.
5. Connect the thermistor connector to the wiring harness connector.
6. Turn the ignition switch to on position.
7. Turn the A/C switch on.
8. Turn the fan switch on.
9. Immerse the sensor part of thermistor in a container of ice water.
10. Verify that magnetic clutch operate as shown below.
 - If not as specified, replace the thermistor.

Water temperature	Differential	Magnetic clutch
0—3°C {32—37°F}	3—5°C {38—41°F}	OFF
13—17°C {34—62°F}	2—6°C {36—42°F}	OFF

CONTROL SYSTEM

REFRIGERANT PRESSURE SWITCH REMOVAL/INSTALLATION

dcf074061503w01

1. Disconnect the negative battery cable.
2. Discharge the refrigerant from the system. (See 07-10-3 RECOVERY.) (See 07-10-3 REFRIGERANT CHARGING.)

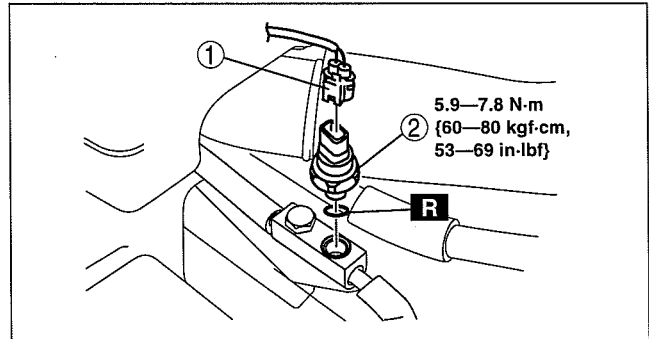
Caution

- If moisture or foreign material enters the refrigeration cycle, cooling ability will be lowered and abnormal noise will occur. Always immediately plug all open fittings after removing any refrigeration cycle parts to keep moisture or foreign material out of the cycle.

3. Remove in the order indicated in the table.

1	Refrigerant pressure switch connector
2	Refrigerant pressure switch

4. Install in the reverse order of removal.
5. Perform the refrigerant system performance test. (See 07-10-2 REFRIGERANT SYSTEM PERFORMANCE TEST.)



DBG740ZWB010

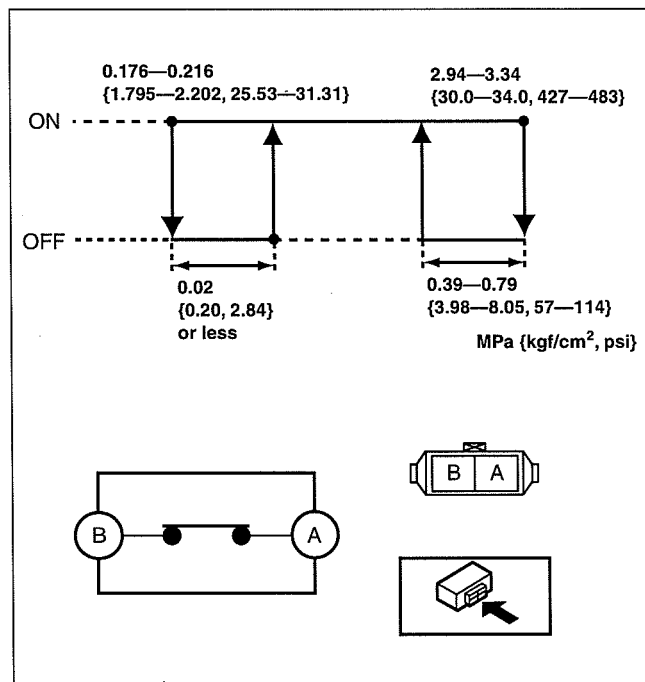
Refrigerant Pressure Switch Installation Note

1. Apply compressor oil to O-ring and connect the joint.

REFRIGERANT PRESSURE SWITCH INSPECTION

dcf074061503w02

1. Install the SST (gas charging set).
2. Disconnect the refrigerant pressure switch connector.
3. Verify the high-pressure side reading of the SST (manifold gauge) and continuity between the refrigerant pressure switch terminals.
 - If there is any malfunction, replace the refrigerant pressure switch.



DBG740ZWB011

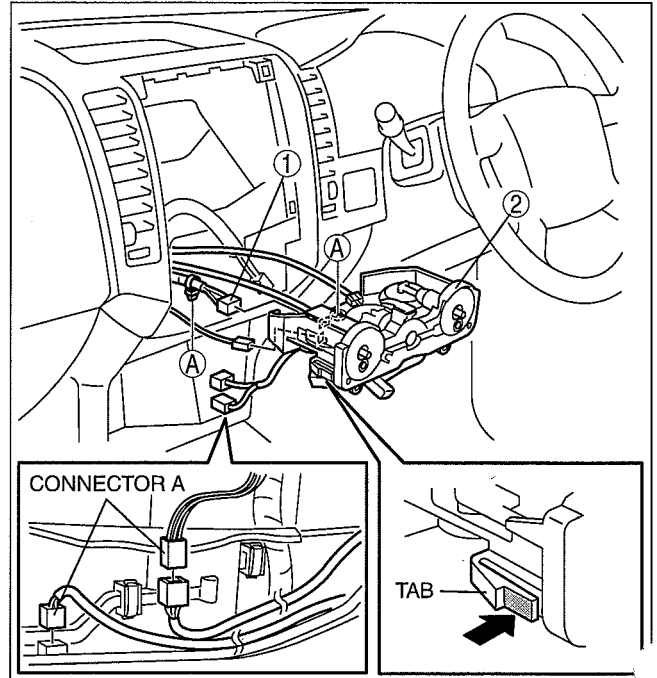
CONTROL SYSTEM

CLIMATE CONTROL UNIT REMOVAL

dcf074061190w01

1. Disconnect the negative battery cable.
2. Remove the glove compartment. (See 09-17-9 GLOVE COMPARTMENT REMOVAL/INSTALLATION.)
3. Remove the center panel module or audio unit. (See 09-20-3 CENTER PANEL UNIT REMOVAL/INSTALLATION [TYPE A].) (See 09-20-5 AUDIO UNIT REMOVAL/INSTALLATION [TYPE B].)
4. Disconnect the connector A.
5. Disconnect the air intake wire from the blower unit.
6. Disconnect the airflow mode wire from the air outlet duct.
7. Release the left and right tabs and remove in the order indicated in the table.

1	Connector B
2	Climate control unit

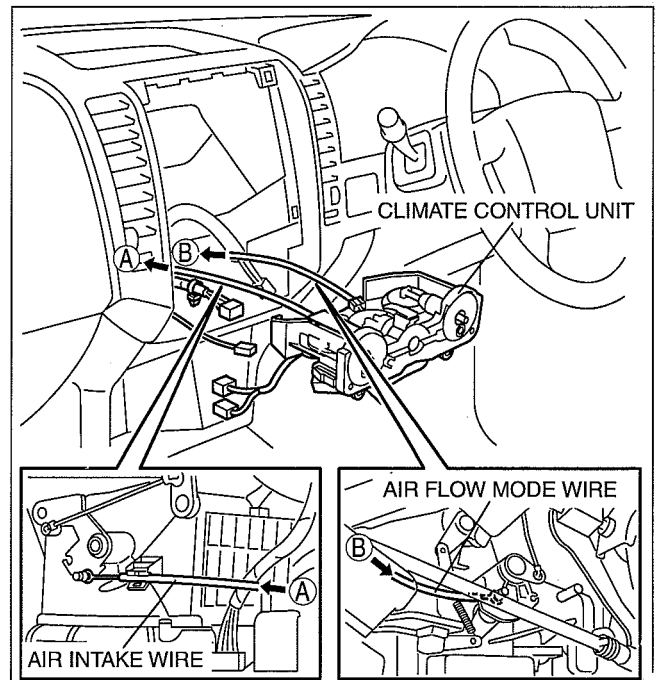


DBG740ZWB034

CLIMATE CONTROL UNIT INSTALLATION

dcf074061190w02

1. Connect the climate control unit connectors.
2. Insert the climate control unit until each clip clicks.
3. Adjust the climate control unit wire. (See 07-40-8 CLIMATE CONTROL UNIT WIRE ADJUSTMENT.)
4. Install the center panel module or audio unit. (See 09-20-3 CENTER PANEL UNIT REMOVAL/INSTALLATION [TYPE A].) (See 09-20-5 AUDIO UNIT REMOVAL/INSTALLATION [TYPE B].)
5. Install the glove compartment. (See 09-17-9 GLOVE COMPARTMENT REMOVAL/INSTALLATION.)
6. Connect the negative battery cable.



DBG740ZWB0

CONTROL SYSTEM

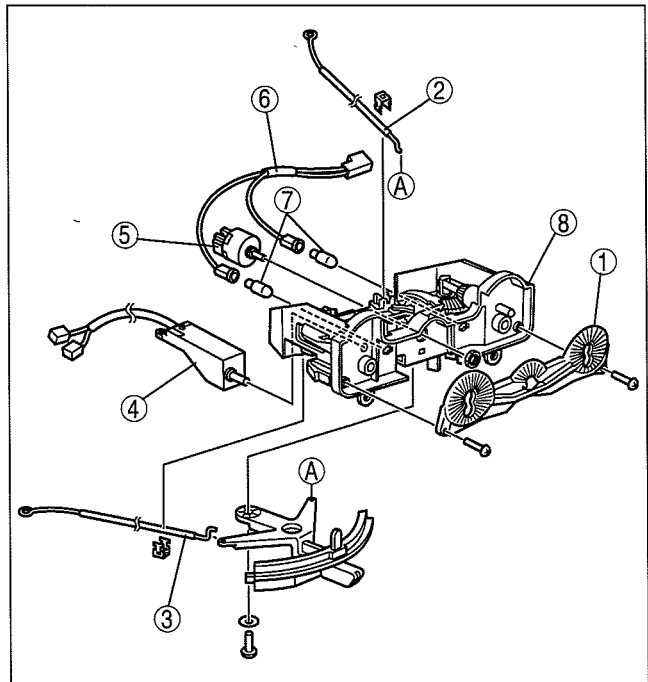
CLIMATE CONTROL UNIT DISASSEMBLY/ASSEMBLY

dcf074061190w03

1. Disassemble in the order indicated in the table.

1	Panel
2	Airflow mode wire
3	Air intake wire
4	A/C amplifier
5	Fan switch
6	Climate control unit harness
7	Illumination bulb
8	Body

2. Assemble in the reverse order of disassembly.



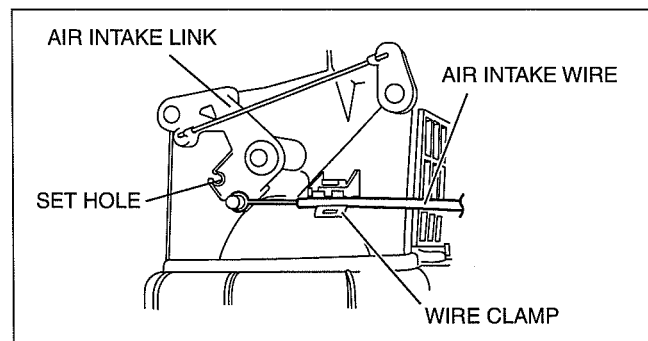
DBG740ZW037

CLIMATE CONTROL UNIT WIRE ADJUSTMENT

dcf074061190w04

Air Intake Wire

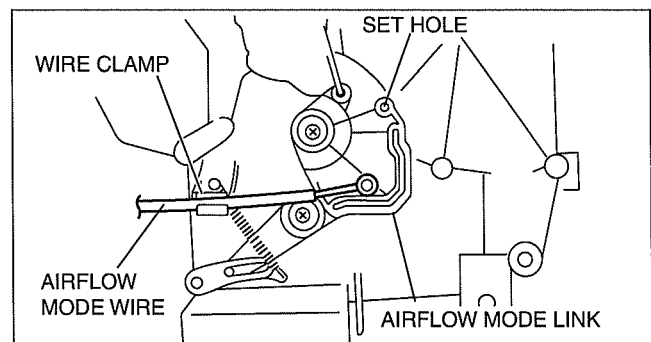
1. Set the REC/FRESH lever at REC.
2. Set the air intake link to REC in the direction of the arrow and insert a screwdriver into the set hole.
3. Connect the air intake wire to the air intake link.
4. Clamp the air intake wire to wire clamp.
5. Verify that the REC/FRESH lever moves its full stroke.



DBG740ZW041

Airflow Mode Wire

1. Set the airflow mode control dial at DEFROSTER.
2. Set the airflow mode link to DEFROSTER in the direction of the arrow and insert a screwdriver into the set hole.
3. Connect the airflow mode wire to the airflow mode link.
4. Clamp the airflow mode wire to wire clamp.
5. Verify that the airflow mode control dial moves its full stroke.



DBG740ZW042

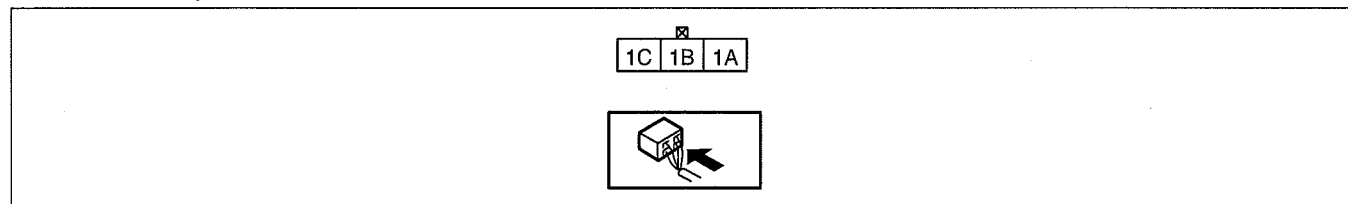
CONTROL SYSTEM

A/C AMPLIFIER INSPECTION

dcf074061190w05

1. Connect the A/C amplifier connector.
2. Turn the engine switch to the ON position.
3. Connect the negative (-) lead of the tester to body ground.
4. By inserting the positive (+) lead of the tester into each A/C amplifier terminal, measure the voltage according to the terminal voltage table.
 - If there is any malfunction, inspect the parts under "Inspection item (s)".
 - If the parts under "Inspection item (s)", replace the A/C amplifier.

Terminal Voltage Table

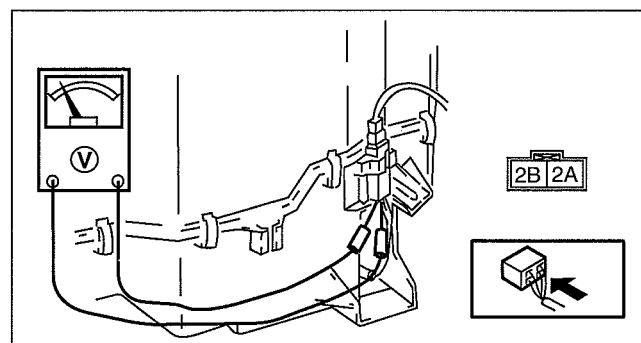


DBG740ZWB699

Term inal	Signal name	Connected to	Measurement condition	Voltage (V)	Inspection item (s)
1A	IG2	A/C2 10 A fuse	Engine switch at ON position	B+	- Inspect for continuity or short circuit (A/C amplifier—A/C2 10 A fuse:1A—A/C2 10 A fuse) - Inspect A/C2 10 A fuse
			Engine switch at LOCK position	Below 1.0	Inspect short circuit (A/C amplifier—A/C2 10 A fuse:1A—A/C2 10 A fuse)
1B	A/C	A/C relay	A/C switch OFF	B+	Inspect or continuity or short circuit (A/C amplifier—A/C relay:1B—Inspect A/C relay)
			A/C switch ON	Below 1.0	Inspect terminal voltage of A/C amplifier (1A)
1C	GND	Ground	Under any condition	Below 1.0	Inspect for continuity (A/C amplifier—ground:1C—GND)

Thermistor Side

1. Connect the thermistor connector.
2. Turn the engine switch to the ON position.
3. Turn the fan switch ON.
4. Turn the A/C switch ON.
5. Connect the positive (+) and negative (-) probes of the voltmeter to terminal 2A and terminal 2B of the thermistor respectively. (The wiring harness connector must be connected to the thermistor connector.)
6. If there is any malfunction, replace the A/C amplifier.



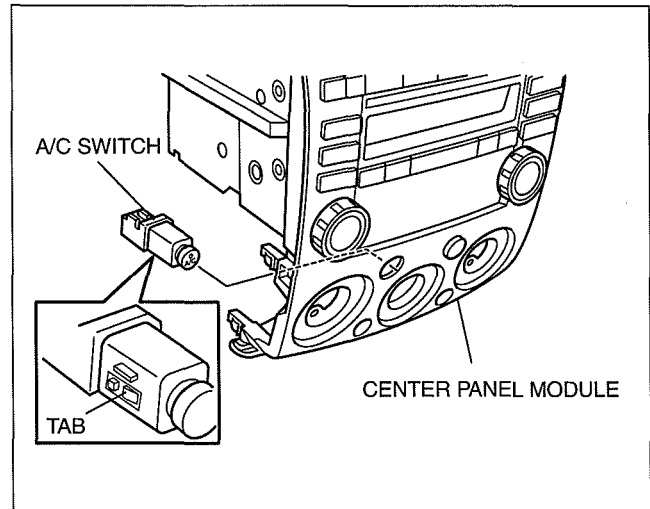
DBG740ZWB600

CONTROL SYSTEM

A/C SWITCH REMOVAL/INSTALLATION

dcf074061190w06

1. Disconnect the negative battery cable.
2. Remove the center panel module or audio unit. (See 09-20-3 CENTER PANEL UNIT REMOVAL/INSTALLATION [TYPE A].) (See 09-20-5 AUDIO UNIT REMOVAL/INSTALLATION [TYPE B].)
3. Disconnect the A/C switch connector.
4. Release the left and right tabs and remove the A/C switch as shown in the figure.
5. Install in the reverse order of removal.



DBG740ZWB045

A/C SWITCH INSPECTION

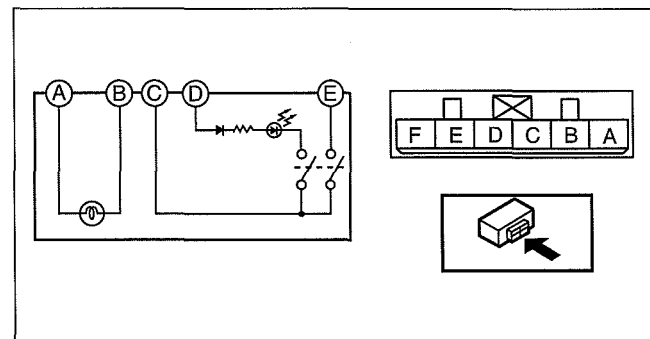
dcf074061190w07

1. Inspect for continuity between the A/C switch terminals using a tester.

○—○ : Bulb ○—○ : Continuity

Switch position	Terminal				
	A	B	C	D	E
OFF	○—○	○—○			
ON	○—○	○—○	○—○	○—○	○—○

DBG740ZWB044



DBG740ZWB043

2. Connect battery positive voltage to terminal D and ground to the terminal C.
3. Turn the A/C switch on.
4. Verify that the LED illuminates.
 - If there is any malfunction, replace the A/C switch.

FAN SWITCH INSPECTION

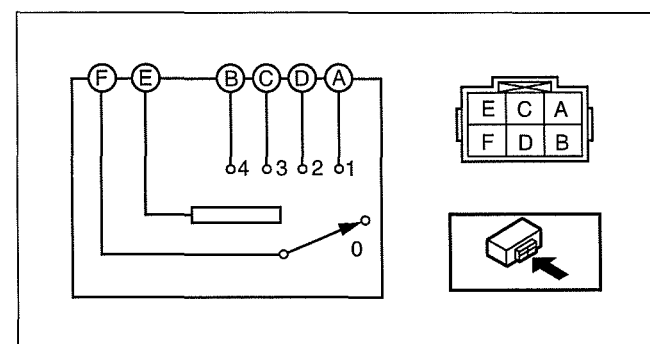
dcf074061050w01

1. Remove the climate control unit. (See 07-40-7 CLIMATE CONTROL UNIT REMOVAL.) (See 07-40-7 CLIMATE CONTROL UNIT INSTALLATION.)
2. Verify that the continuity between the fan switch terminals is as indicated in the table.
 - If there is any malfunction, replace the fan switch.

○—○ : Continuity

Switch position	Terminal					
	A	B	C	D	E	F
0						
1	○—○					○—○
2				○—○		○—○
3			○—○		○—○	○—○
4		○—○			○—○	○—○

DBG740ZWB025



DBG740ZWB026

TECHNICAL DATA

07-50 TECHNICAL DATA

HVAC TECHNICAL DATA 07-50-1

HVAC TECHNICAL DATA

dcf075000000w01

Item			Specification
REFRIGERANT SYSTEM			
Refrigerant	Type		R-134a
	Regular amount (approx. quantity) (g {oz})		550 {19.4}
BASIC SYSTEM			
A/C compressor	Lubrication oil	Type	FD46XG
		Sealed volume (approx. quantity) (ml {cc, fl oz})	180 {180, 6.08}
CONTROL SYSTEM			
A/C compressor	Magnetic clutch clearance (mm {in})		0.35—0.75 {0.014—0.029}

(6)

(7)

(8)

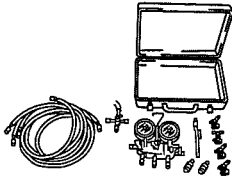
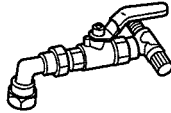
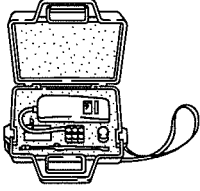
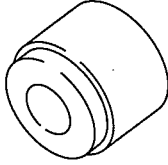
SERVICE TOOLS

07-60 SERVICE TOOLS

HEATER, VENTILATION &
AIR CONDITIONING (HVAC) SST 07-60-1

HEATER, VENTILATION & AIR CONDITIONING (HVAC) SST

dcf07600000w01

49 C061 0A0B Gas Charge Set 	49 C061 012 Anti-back flow valve 	49 C061 013 Gas Leak Tester 
49 H061 005 Attachment 	—	—

